
When Outlier Supervision Fails: Self-Supervised Robustness in Deep Structure-from-Motion

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Abstract

To reconstruct cameras and 3D structure from real image collections, learning-based Structure-from-Motion must reject outliers: wrong correspondences that survive geometric verification. The standard recipe trains an outlier classifier with labels derived from an already-trusted reconstruction of the training scenes — supervision that presupposes exactly the capability it is meant to provide. We ask when this recipe fails, and show its failures are caused by the labels themselves, not by domain shift: labels cannot be manufactured where robustness is needed most, and retraining the state-of-the-art supervised method (RESfM) on a 60%-contaminated domain with best-case labels still loses $2\times$ to a label-free alternative. That alternative is SS-RESfM: RESfM’s outlier head trained with a self-supervised loss — confident pseudo-labels formed from percentiles of the model’s own reprojection error — with no ground-truth or COLMAP labels anywhere in training or test. Under a strictly controlled protocol (identical tracks, robust bundle adjustment, and training schedule, varying only the outlier mechanism) we compare label-free statistical removal (madweight) and soft reweighting against a from-scratch RESfM baseline across six datasets spanning 0.5%–61% measured outliers. The optimal mechanism flips with contamination. Removal wins the high-contamination regime: at 61% plain removal beats the supervised baseline on every seed pair (12.4 ± 2.1 vs. 19.2 ± 3.4) and both beat COLMAP, while at 43% it beats every supervised seed’s median ($1.4\text{--}1.9$ vs. $2.4\text{--}4.3$). Reweighting wins the low-contamination regime, beating the supervised baseline $2.5\text{--}2.9\times$ on two of the three clean out-of-distribution splits (BlendedMVS and Olsson; Strecha goes to supervision on every seed) — and, unlike any supervised recipe we test, does so reliably: the label-free arms land in a narrow band on every training seed, whereas the supervised baseline collapses on the clean benchmark in most seeds under every learning rate, budget, and recipe variant we try (on Olsson — 39 essentially outlier-free scenes, 0.5% — every MegaDepth-trained supervised variant lands at a translation error of 7.4–9.1, versus 2.9–3.6 for every label-free arm). Supervision retains the in-distribution regime and the cleanest OOD mean (self-supervised arms within $\sim 38\%$ in distribution). Every comparison is backed by five-training-seed bands, a paired-seed evaluation control, and an environment-noise study — in a benchmark family where prior work fixes a single seed. We trace this to a structural circularity: outlier labels require a trusted reconstruction of the very scenes whose contamination makes reconstruction hard (for any gold-standard labeler, not just COLMAP), and we measure the resulting label ceiling: regenerating supervision from scratch, label recall collapses from 0.82 to ~ 0.3 above $\sim 15\%$ contamination, precisely where robustness is needed. Label-free adaptivity is thus a missing ingredient for robust deep SfM.

1 Introduction

Structure-from-Motion (SfM) recovers camera poses and 3D structure from unordered image collections. Its correspondences are heavily contaminated by outliers (wrong matches that survive pairwise geometric verification), particularly in Internet photo collections where repeated and symmetric structure produces spurious epipolar geometries. Classical pipelines reject outliers with RANSAC and robust bundle adjustment (BA); recent deep equivariant methods (ESfM (Moran et al., 2021), RESfM (Khatib et al., 2025)) instead learn a sets-of-sets permutation-equivariant network that regresses cameras from point tracks and, in RESfM, adds a supervised per-point inlier/outlier classification module. RESfM’s labels are derived from a COLMAP (Schönbberger and Frahm, 2016) reconstruction in two stages: first by track membership, then refined by a 4-pixel reprojection test against COLMAP-pose triangulation (Khatib et al., 2025).

We take a different route. Rather than supervise the head with COLMAP labels, SS-RESfM trains it self-supervised: an adaptive confidence-weighted loss forms confident pseudo-labels from percentiles of the model’s own reprojection error, so no ground-truth or COLMAP outlier labels are used anywhere in training or test. On top of this self-supervised head we study a question RESfM’s single supervised mechanism leaves open: is one outlier-handling mechanism optimal across the range of contamination levels SfM encounters? We find that it is not. Holding everything else fixed and varying only the outlier-handling mechanism, the best mechanism flips with contamination: hard removal at high contamination, soft reweighting at low, with the supervised classifier retaining the typical in-distribution scene and the cleanest OOD mean (Fig. 1). We further show the removal family wins the high-contamination regime outright: plain removal beats the supervised baseline on every seed at both 43% and $\sim 60\%$, while madweight, which discards points by the median-absolute-deviation (MAD) of their reprojection error and soft-weights the survivors with no learned threshold, matches the baseline there and beats COLMAP. We attribute this to (i) domain shift, since the classifier is trained only on in-distribution MegaDepth, and (ii) supervised labeling’s dependence on a COLMAP reconstruction and ground-truth poses, readily obtained for curated data but not for the contaminated collections where robustness is most needed; a label-free, per-scene mechanism is attractive precisely in that regime.

The appeal of self-supervision here is structural, not incidental. Outlier labels must be derived from a trusted reconstruction of the very scenes whose contamination makes reconstruction hard, so supervision bootstraps from precisely the capability it is meant to provide. This circularity is not specific to COLMAP: any gold-standard labeler must first solve the scene in order to label its outliers, and it is exactly the challenging, contaminated collections that defeat this bootstrapping. A self-supervised head, by contrast, consumes only the point tracks themselves; the training data for robust SfM then becomes as unlimited as unlabeled imagery, and new domains require no labeling infrastructure at all.

Contributions. (1) The label ceiling, measured and shown causal: outlier supervision requires a trusted reconstruction of the very scenes whose contamination makes reconstruction hard. We measure the resulting label collapse (recall $0.82 \rightarrow \sim 0.3$ above $\sim 15\%$ contamination) and prove it is causal, not domain shift: retrained on a 60%-contaminated domain with best-case labels, supervision still loses $2\times$ to label-free removal (§5.5). (2) The complementarity finding: under an identical tracks+BA+schedule pipeline isolating the mechanism alone, the optimal outlier mechanism flips with contamination — removal at high contamination (beating the supervised baseline on every seed pair at $\sim 60\%$), soft reweighting on clean OOD data ($2.5\text{--}2.9\times$, seed-reliably) — with supervision retaining a curated middle band. (3) SS-RESfM, the instrument of the study and a constructive answer: RESfM’s outlier handling made entirely self-supervised via confident pseudo-labels from the model’s own reprojection-error percentiles — no ground-truth or COLMAP labels anywhere — matching the supervised baseline where labels are sound and surpassing it where they are not; plus madweight, which needs not even the head, removing by robust reprojection statistics (MAD) and beating COLMAP where COLMAP’s own labels cannot be manufactured. (4) A seed- and environment-level reliability methodology for post-BA SfM benchmarks: five-training-seed bands for every cell, a paired-seed evaluation control (2,352 evaluations showing the evaluation seed is immaterial), an environment-noise floor for post-BA means,

and per-seed-pairing claim verification — practices absent from this benchmark family, where prior work fixes a single seed, and which surface findings (bimodal clean-ODD failures, basin-dependent escapes) that mean $\pm$ std alone would hide. (5) A reusable robustness benchmark: rebuilt Appendix-C-protocol track suites for five OOD datasets with measured contamination (0.5–61%), GT-derived outlier labels, per-domain train/test splits, and the label-quality benchmark — released with code and configurations.

2 Related Work

Deep equivariant SfM. Learning-based SfM regresses cameras from point tracks with permutation-equivariant networks. ESfM (Moran et al., 2021) casts uncalibrated SfM as deep projective factorization over a measurement tensor, trained with an unsupervised reprojection loss; GASfM (Brynte et al., 2024) uses a graph-attention variant. Both assume outliers were already removed by two-view geometric verification. RESfM (Khatib et al., 2025) targets contaminated internet collections with a sets-of-sets architecture plus a supervised per-point inlier/outlier classification module and a robust BA, reporting accuracy close to its own clean-track upper bound (“ESfM*”) and on par with classical Theia/GLOMAP (often with better translation recovery, while registering fewer images). End-to-end pointmap and pose-regression pipelines (VGGSfM, MAST3R, DUST3R) target far smaller image sets. Concurrently with this work, the RESfM authors moved robustness to the pose-averaging stage: VGPA (Khatib et al., 2026) learns globally consistent camera poses from noisy view-graphs with a self-supervised relative-pose consistency objective. That development independently corroborates our thesis that label-free robustness is the frontier of deep SfM, and is complementary in stage and input modality: VGPA consumes pairwise relative poses (requiring a two-view front-end and no track structure), whereas we operate on point-track matrices with per-observation outliers and recover structure jointly; direct numeric comparison across the two paradigms is confounded for the same reason RESfM does not compare head-to-head with pose-averaging pipelines. Our contribution is orthogonal to both: we propose no new architecture — we reuse the RESfM/ESfM backbone and study the outlier-handling stage and its supervision in isolation, with a seed- and environment-level reliability methodology that applies to either paradigm.

Outlier rejection. Correspondence outliers that survive RANSAC/two-view filtering are the central difficulty in internet-scale SfM. Classical pipelines suppress them with M-estimators and robust BA; RESfM casts rejection as supervised classification. We contrast supervised classification against label-free statistical removal via the median absolute deviation (MAD) (Rousseeuw and Croux, 1993) and soft reweighting, and show their relative merit is contamination-dependent.

Robust bundle adjustment. Following (Khatib et al., 2025), we apply a robust BA that removes high-reprojection and low-visibility points, keeps the largest connected component, re-triangulates, and re-optimizes. This stage is identical across all our arms.

Test-time adaptation. Per-scene fine-tuning at inference is used by ESfM/ RESfM to reach sub-pixel accuracy. We treat it as an orthogonal option ($\pm$ TTT) applied uniformly across mechanisms.

3 Method

3.1 Background

The input is a (sparse) point-track tensor M of m tracks across n cameras, built by concatenating pairwise RANSAC-verified SIFT matches. The sets-of-sets permutation-equivariant network (Khatib et al., 2025; Moran et al., 2021) maps M to per-camera poses and a per-point outlier score $s \in [0, 1]$ (points in the same track may receive different scores). Crucially, unlike RESfM, whose classification head is trained with a BCE loss against COLMAP-derived inlier/outlier labels, SS-RESfM trains this head self-supervised, with an adaptive confidence-weighted loss: per forward pass it forms confident pseudo-labels from percentiles of the model’s own reprojection error (lowest $p\%$ inliers, highest $p\%$ outliers) and applies

Table 1: Datasets: outlier rates (RESfM-reported vs. measured on our tracks) and regime.

Dataset	RESfM% ours%		Regime
Strecha (Strecha et al., 2008)	2.7	1.7	clean, LIDAR GT
BlendedMVS (Yao et al., 2020)	3.6	3.1	low-outlier
MegaDepth (Li and Snavely, 2018)	25.6	25.4	contam., in-dist.
1DSfM (Wilson and Snavely, 2014)	31.9	43.3	contam., OOD
1DSfM-hard	–	~60	extreme, OOD
Olsson (Olsson and Enqvist, 2011)	–	0.5	outlier-free

BCE only to those. No ground-truth or COLMAP outlier labels are used at any point. A reprojection error e is available for every observation once an initial reconstruction exists. All mechanisms below consume s (and, for madweight, e), produce a per-observation weight w (or a hard keep/drop decision), and feed the result to the identical robust-BA stage (Fig. 2).

3.2 Outlier-handling mechanisms

We compare four mechanisms, illustrated as weight-vs-signal transfer functions in Fig. 2:

- remove (the RESfM analogue): drop a point if $s > \tau$; keep with $w=1$ otherwise. We use RESfM’s default $\tau=0.6$ and its $\tau=0.8$ for the low-outlier, single-camera Strecha/BlendedMVS sets (Khatib et al., 2025).
- weight: soft, $w = 1 - s$, no removal.
- hybrid: three-band: keep (s below a low percentile), soft-weight (middle), drop (s above a high percentile).
- madweight (ours): removal is driven by robust reprojection statistics, not the head. An observation is removed if its reprojection error exceeds $\text{med}(e) + \alpha \cdot \text{MAD}(e)$ ($\alpha=2$); surviving observations are soft-weighted by $w = 1 - s$. madweight is the only mechanism that consumes the reprojection branch e and requires no learned threshold for removal.

Each mechanism is evaluated with and without test-time fine-tuning ($\pm\text{TTT}$), in which the head continues to adapt on the test scene.

3.3 Test-time fine-tuning and robust BA

We load a multi-scene checkpoint, run a per-scene forward pass to obtain s and e , apply the mechanism, fine-tune per scene (~ 1000 iterations) under the reprojection objective, and apply robust BA. This protocol, the seed, and the schedule are held fixed across all arms; only the mechanism block changes.

4 Experiments

4.1 Datasets and contamination

We evaluate on six datasets spanning a wide contamination range. Alongside the RESfM paper’s quoted outlier fractions we report the contamination measured on our own tracks (robust ground-truth-pose triangulation with the standard 4px cutoff, applied uniformly to every dataset); see Table 1. On MegaDepth (where tracks are identical to RESfM’s) the two measurements agree (25.4 vs 25.6), validating our measurement; our 1DSfM rebuild is markedly harder than the tracks behind RESfM’s quoted 31.9%, which should be kept in mind when reading the 1DSfM columns. This difference also yields a falsifiable prediction: since the winning mechanism is a function of contamination (§5.5), the supervised advantage should persist on lower-contamination track builds — in particular, on the authors’ own $\sim 32\%$ 1DSfM tracks we expect supervision to be competitive or ahead, with the label-free advantage emerging as contamination approaches our rebuild’s 43% and

dominating by 60%. We state this in advance of evaluating on those tracks. 1DSfM-hard comprises the five 1DSfM scenes unused by the standard ten-scene split (Gendarmenmarkt, Piccadilly, Roman Forum, Trafalgar, Union Square), built with the same pipeline and sub-sampled to 300 cameras per scene following RESfM’s convention. Networks are trained on 27 MegaDepth scenes; MegaDepth test is therefore in-distribution, while 1DSfM/1DSfM-hard/Strecha/BlendedMVS/Olsson are cross-dataset (out-of-distribution).

4.2 Protocol, baselines, and track provenance

Following RESfM’s public training code and default configuration, all networks train for 20,000 epochs with seed 20 (the seed reported by RESfM (Khatib et al., 2025)) as the primary seed; the self-supervised arms are additionally replicated with seeds 21/22, and cells marked [‡] report mean $\pm$ std over the three seeds, including the supervised baseline, whose own seed band is reported throughout (its MegaDepth mean is notably seed-volatile, 0.32 ± 0.12 , while its median is stable, 0.06 ± 0.01). Batch size, image subsampling, the eleven-milestone learning-rate decay schedule (10k, 11k, ..., 20k, $\gamma=0.9$), and validation-based early stopping follow that released configuration and are held identical across all arms. Two deliberate settings are disclosed. First, the base learning rate is per-method: 10^{-3} for the supervised arms (RESfM’s released value) and 10^{-4} for the self-supervised arms, whose adaptive pseudo-label loss converges early (best checkpoint by epoch ~ 7.5 k, before the first decay milestone; the supervised arms peak at epoch ~ 16.5 k); both are well within the budget, so neither method is truncation-limited. Second, two secondary values (the infinity-point margin and the validation metric used for checkpoint selection) are held identical across all arms rather than at RESfM’s released values, preserving the controlled comparison. The selection deviation demonstrably favors the baseline: under the authors’ own Accuracy-based checkpoint selection the supervised baseline is weaker on every dataset (RESfM-faithful row in Table 2; MegaDepth 0.50 vs. 0.34), so all reported supervised-baseline margins are lower bounds, and the baseline’s clean-benchmark failures persist under its own selection rule (Olsson 8.8, BlendedMVS rotation 53°). We report two architectures: shallow (1×3) and deep (2×3).

Fair-comparison anchor. Our controlled baseline is RESfM-from-scratch: trained by us with the identical setup as our arms (same 27 scenes, sampling, epochs, seed), so only the outlier mechanism differs. We additionally report RESfM-released (authors’ weights) and RESfM (paper).¹

Track provenance. MegaDepth tracks, preprocessing, and the 36 test scenes are identical to RESfM, so MegaDepth is directly comparable to published numbers. The OOD tracks (1DSfM/1DSfM-hard/Strecha/BlendedMVS/Olsson) are built by us from the publicly released images and ground truth (feature extraction, matching, and track formation with our pipeline; RESfM did not release OOD tracks); OOD comparisons are therefore controlled within our pipeline and are not directly comparable to RESfM’s published OOD numbers. On MegaDepth, RESfM’s own released weights yield 0.209 in our harness (vs. their reported 0.169), confirming our evaluation is faithful and that ~ 0.209 is the reproducible level; this is unchanged (0.217) under the authors’ exact pinned software stack (MKL BLAS, same pycolmap/pyceres), isolating the residual gap to solver/hardware-level nondeterminism in robust BA (App. G). All reported results come from this single pipeline and software stack; a stack-replication study (App. G) shows the no-TTT arms are insensitive to it, while test-time fine-tuned means inherit solver-level nondeterminism and should be read with that variance in mind.

Schedule robustness. All reported results use RESfM’s released eleven-milestone schedule. An earlier accidental five-milestone variant of ours was retired; a three-seed replication of both schedules shows the schedule effect lies within seed noise (MegaDepth 0.46 ± 0.08 vs. 0.40 ± 0.09 ; 1DSfM 9.6 ± 3.2 vs. 12.2 ± 2.4 , per-seed rankings inverting on both), with mechanism orderings stable throughout.

¹The ESfM row likewise follows the unified 20k-epoch protocol. ESfM’s own released multi-scene (calibrated) recipe trains for 30k epochs plus a 500-epoch per-scene fine-tune; we deliberately hold the schedule fixed across all rows so that the outlier mechanism remains the only controlled variable.



Figure 1: The optimal outlier mechanism flips with contamination. Translation (left) and rotation (right) means (log scale) of the self-supervised arms vs. supervised RESfM, five of the six datasets ordered by measured contamination (Table 2 data). Circled = best per dataset. The rotation panel mirrors the translation ordering, with a self-supervised arm best on 1DSfM and the supervised BlendedMVS failure even more pronounced (36°).

A method-isolated comparison, not a SOTA claim. Because RESfM-from-scratch shares every pipeline component with our arms and differs only in the outlier mechanism, it isolates the effect of removing supervision, the same controlled protocol RESfM itself uses when it re-trains ESfM/GASfM in its own pipeline. The anchor is also authors-faithful: five retrains under the authors’ exact validation metric and margin reproduce our declared baseline’s five-seed band on every dataset (App. L), so the same baseline is simultaneously recipe-faithful and protocol-controlled — the comparison does not depend on which supervised recipe one prefers. We do not claim state of the art: RESfM’s released checkpoint (0.209 on MegaDepth) is stronger than either from-scratch arm. This reproduction gap stems from the released weights themselves, not our pipeline (the authors’ own training code on our tracks reproduces 0.34–0.39, not 0.209), so it handicaps the supervised and self-supervised arms equally. We report RESfM-released and RESfM (paper) as reference rows throughout.

Metrics. Following RESfM’s convention we draw conclusions from mean errors (translation emphasized, rotation reported alongside), post-BA (rotation in degrees; translation is the aligned camera-position error in scene coordinates), and provide medians as supporting statistics (RESfM likewise reports medians in its appendix). The one declared exception is 1DSfM, whose mean is dominated by two symmetric-structure failure scenes (Ellis Island, Tower of London) that fail for all methods; there we treat the median as the representative figure. Rotation and registered-camera counts (N_r) for all arms are tabulated in the supplementary (appendix); rotation broadly agrees with translation (the supervised BlendedMVS failure is even larger in rotation), and under high contamination the label-free arms register more cameras than the supervised baseline (38–43% vs. 37% of ground-truth cameras on 1DSfM-hard). Two registration notions must be distinguished: every learned arm outputs a pose for every camera (pose coverage; COLMAP genuinely fails on 36% of 1DSfM-hard), while N_r counts the cameras surviving robust BA and a 5-pixel quality mask, the set over which errors are computed. Under this stricter notion the plain-removal arms attain their low errors over a reduced set (59%/28% vs. the baseline’s 77%/37% on 1DSfM/1DSfM-hard), consistent with aggressive pruning disconnecting weakly-supported cameras; madweight is registration-matched to the baseline (74%/38%), so its comparisons are not coverage artifacts.

5 Results

5.1 Main comparison

Table 2 is the controlled head-to-head under the RESfM-aligned schedule: every row shares tracks, architecture, schedule, robust BA, and seed, so any difference is attributable to

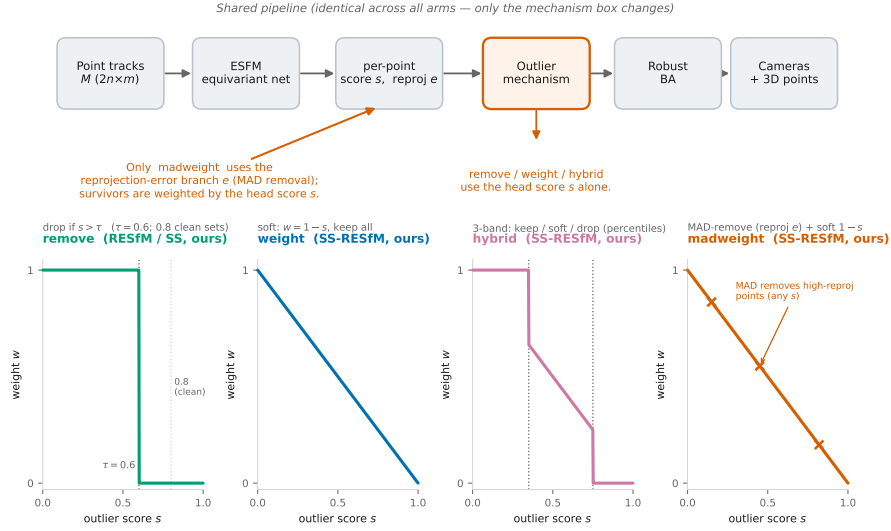


Figure 2: Method. A shared pipeline (top) in which only the outlier-mechanism block changes. The four mechanisms as weight-vs-signal transfer functions (bottom). Only madweight consumes the reprojection-error branch e , and only for its removal step; its surviving points are still weighted by the head score s ($w=1-s$). The other three mechanisms use s alone.

the outlier mechanism and its supervision alone. Reading along the contamination axis:

(1) Clean OOD (Olsson 0.5%, Strecha 1.7%, BlendedMVS 3.1%): the label-free arms win decisively where the supervised head misfires, and the wins are seed-robust on both sides: every arm beats the full baseline band on BlendedMVS at five seeds each ($0.14-0.24$ vs. 0.35 ± 0.04 , and the soft arms at 0.14 ± 0.02 are $2.5\times$ better; the baseline’s rotation failure, 31.7° averaged over seeds, persists on every one) and by $\sim 3\times$ on Olsson means against four of five baseline seeds (3.0 ± 0.3 vs. $8.2-8.9$; one baseline seed escapes the collapse at 2.69 , revealing the clean-benchmark failure as basin-dependent rather than universal, its frequency $4/5$). Figure 3 shows what the collapse looks like: on an essentially outlier-free scene, the supervised reconstruction degenerates to a compressed cluster with 16.5° mean rotation error, while the label-free arm reproduces the ground-truth camera arc nearly exactly. The collapse is not a from-scratch artifact: the authors’ released checkpoint also collapses on Olsson (9.10; identically, 9.18, under their pinned software stack), as does the authors-faithful configuration (8.77). Strecha is the exception: RESfM’s mean is better on every seed ($0.006-1.67$ vs. ~ 2 , though one baseline seed nearly loses it) and at seed 20 the best label-free median ties it (0.005 , Table 6). (2) In distribution (MegaDepth), supervised RESfM retains mean and median. Its mean is strongly seed-dependent (0.37 ± 0.12 over five seeds; its best, 0.19 , approaches the released checkpoint) while its median is stable (0.07 ± 0.02); the deterministic self-supervised arms band at 0.51 ± 0.08 and the TTT arm at 0.40 ± 0.11 over five seeds; a $\sim 38\%$ band-level gap without labels, not a win (against the authors-faithful configuration, 0.496 , the arms reach parity, and the TTT arm is ahead). Medians tell the same story more tightly ($0.13-0.20$ vs. 0.07). (3) Moderate contamination (1DSfM, 43%) has no stable winner between madweight (11.0 ± 1.7) and the baseline (10.5 ± 1.5 ; overlapping bands), so we decline a directional claim for madweight. The removal family leads on both statistics in band terms: remove+TTT 8.9 ± 1.2 vs. the baseline’s 10.5 ± 1.5 (separated means, overlapping tails), and plain remove’s medians (1.7 ± 0.2) beat every baseline seed pairwise at five seeds each ($1.4-1.9$ vs. $2.4-4.3$; its means, 9.1 ± 2.2 , remain failure-scene-dominated): a label-free win at 43% (Table 6), established against the baseline’s own five-seed band. (4) Extreme contamination (1DSfM-hard, $\sim 60\%$): removal beats weighting robustly, madweight (18.9 ± 4.1) statistically matches the supervised baseline (19.2 ± 3.4), and plain remove goes further: 12.4 ± 2.1 over five seeds, below the baseline on every one of the 25 seed pairs (its worst seed, 15.8 , beats the baseline’s best, 16.2), with tight medians (10.8 ± 2.8

Table 2: Main controlled comparison (RESfM-aligned schedule). Mean / median translation error (post-BA), seed 20. All rows share tracks, architecture, schedule, robust BA, and seed; only the outlier mechanism and its supervision differ. Reference rows (means) below the rule; upper block = SS-RESfM arms. Best per column in bold. [†]mean $\pm$ std over three evaluation repeats (TTT is run-to-run nondeterministic; appendix). All self-supervised arms and the supervised baseline are five-seed bands (seeds 20–24, mean $\pm$ std of per-seed dataset means); reference and classical rows are single runs, except [§]RESfM (released), where the band is over five evaluation seeds on the authors’ fixed weights (MegaDepth, 1DSfM, Strecha, BlendedMVS): with the network held constant the pipeline is near-deterministic, which is the control for reading the training-seed bands above (App. L). Lower panels: rotation and registered-camera fraction (metric priority: translation, then rotation, then N_r , following RESfM). ESfM*: vanilla ESfM trained and tested on ground-truth-cleaned tracks (clean-track reference; its mean is failure-scene-dominated, as in RESfM’s own Table 1).

Arm	MegaDepth (25.4) $N_c=13126$	1DSfM (43.3) 4563	1DSfM-hard (60) 1497	Strecha (1.7) 54	BMVS (3.1) 225	Olsson (0.5) 3858
Translation error (mean / median), the primary metric						
madweight	0.51 $\pm$ 0.08/0.16	11.0 $\pm$ 1.7/4.9	18.9 $\pm$ 4.1/13.4	2.65 $\pm$ 0.25/1.16	0.14 $\pm$ 0.02/0.05	3.0 $\pm$ 0.3/1.4
weight	0.51 $\pm$ 0.09/0.20	14.4 $\pm$ 3.0/9.6	22.9 $\pm$ 2.8/15.6	1.97 $\pm$ 0.09/0.04	0.14 $\pm$ 0.03/0.02	2.9 $\pm$ 0.3/1.1
weight+TTT	0.40 $\pm$ 0.11/0.18	15.7 $\pm$ 2.1/10.8	22.2 $\pm$ 3.2/15.3	2.00 $\pm$ 0.05/0.12	0.14 $\pm$ 0.02/0.01	8.0 $\pm$ 0.3/0.6
remove	0.51 $\pm$ 0.08/0.13	9.1 $\pm$ 2.2/1.7 $\pm$ 0.2	12.4 $\pm$ 2.1/10.8	3.20 $\pm$ 0.11/2.03	0.24 $\pm$ 0.06/0.17	3.4 $\pm$ 0.6/1.7
remove+TTT	0.56 $\pm$ 0.07/0.14	8.9 $\pm$ 1.2/2.0	15.2 $\pm$ 1.0/11.3	3.01 $\pm$ 0.07/1.70	0.22 $\pm$ 0.10/0.18	3.6 $\pm$ 0.4/1.9
RESfM-scratch (supervised)	0.37 $\pm$ 0.12/0.07	10.5 $\pm$ 1.5/3.4	19.2 $\pm$ 3.4/16.5	0.39 $\pm$ 0.72/0.01	0.35 $\pm$ 0.04/0.33	7.4 $\pm$ 2.6/0.95
RESfM (released) [§]	0.203 $\pm$ 0.008	10.71 $\pm$ 0.27	15.29	0.20 $\pm$ 0.00	0.33 $\pm$ 0.00	9.10
RESfM (paper)	0.169	–	–	–	–	–
RESfM-faithful (authors’ ckpt selection)	0.496	9.75	17.58	0.144	0.367	8.77
ESfM* (clean tracks)	0.60 $\pm$ 0.18	13.24 $\pm$ 0.52	18.12 $\pm$ 4.35	1.14 $\pm$ 0.45	3.79 $\pm$ 3.51	–
ESfM (same tracks, no outlier mech.)	0.71 $\pm$ 0.11	18.42 $\pm$ 0.75	21.76 $\pm$ 0.80	2.03 $\pm$ 0.14	6.45 $\pm$ 3.75	5.91 $\pm$ 3.32
GASfM (released ckpt, our robust BA)	2.25	25.84	36.49	1.14	11.13	4.45
GLOMAP (classical, global)	3.33	27.21	35.42	0.047	0.339	3.17
COLMAP (classical, increm.)	0.62 $\pm$ 0.03	6.29 $\pm$ 0.42	20.4 $\pm$ 1.1	0.03 $\pm$ 0.00	0.01 $\pm$ 0.00	0.20 $\pm$ 0.00
Rotation error ($^\circ$, five-seed means; classical row single run)						
madweight	3.70	7.74	27.7	12.80	9.01	14.4
weight	3.28	8.24	28.0	7.61	9.17	13.2
weight+TTT	2.88	10.09	30.6	7.95	8.30	11.6
remove	4.04	6.94	12.3	14.39	16.04	13.9
remove+TTT	4.22	6.90	16.4	13.23	13.19	14.5
GLOMAP (classical)	5.30	8.78	26.4	0.25	2.15	1.29
RESfM-scratch (supervised)	1.89	9.62	27.3	1.56	31.66	12.6
Registered cameras N_r/N_c (five-seed means; classical row single run)						
madweight	81%	70%	43%	91%	92%	86%
weight	79%	79%	40%	98%	94%	90%
weight+TTT	77%	81%	41%	99%	94%	90%
remove	72%	58%	29%	97%	88%	78%
remove+TTT	71%	58%	32%	96%	90%	78%
RESfM-scratch (supervised)	85%	77%	34%	99%	92%	89%
GLOMAP (classical)	96%	96%	80%	100%	86%	100%

vs. 16.5 ± 4.8); it also beats the authors’ released checkpoint there (15.29, environment-insensitive), so the extreme-contamination win holds against every supervised variant we can construct or obtain; all label-free arms are far ahead of COLMAP (§5.4). Rotation means (Fig. 1, right) follow the same ordering on MegaDepth/Strecha/BlendedMVS/Olsson (e.g. BlendedMVS 8.3–9.2 $^\circ$ for the soft self-supervised arms vs. 31.7 $^\circ$ supervised) and mildly favor madweight and weight on 1DSfM (7.7–8.2 $^\circ$ vs. 9.6 $^\circ$; weight+TTT trails at 10.1 $^\circ$) even where translation favors the baseline.

Classical baselines. Running GLOMAP (global) and COLMAP (incremental) on the same tracks with known intrinsics reproduces the classical/deep trade-off this literature reports, and sharpens it. GLOMAP is far the best method on pristine calibrated data (Strecha 0.047, an order of magnitude below every learned method; Olsson median 0.087) and registers almost everything (96–100% of cameras). Under contamination its translations collapse (3.3 on MegaDepth, 27.2 at 43%, 35.4 at 60%, vs. 9.1 and 12.4 for label-free removal) while its rotations stay strong (8.9 $^\circ$ at 43%, better than the supervised baseline’s 9.6 $^\circ$). The failure of classical global SfM under outlier contamination is therefore specifically a positioning failure, not a loss of orientation structure, which is also why translation is the metric of record in this setting. The comparison is not compute- or coverage-matched: GLOMAP keeps 96% of cameras at 43% contamination where plain removal keeps 58%, so the learned arms are far more accurate on a smaller retained set.

Buddah Tooth Relic Temple (Olsson, 0.5% contamination), both trained on MegaDepth: the clean-OOD collapse

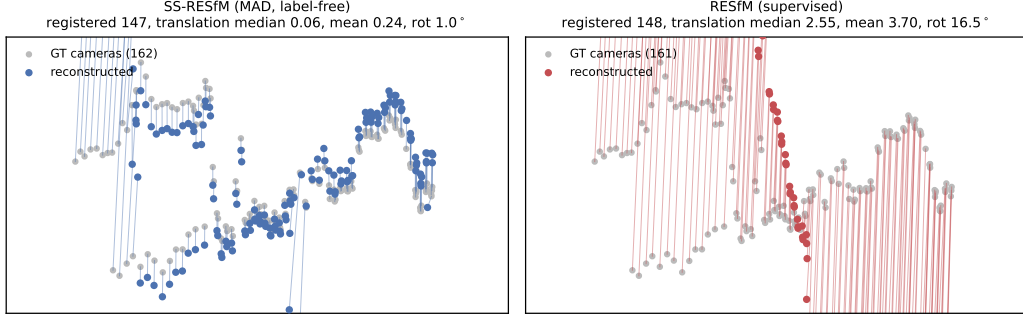


Figure 3: The clean-OOD collapse, qualitatively (Buddah Tooth Relic Temple, Olsson, 0.5% contamination; both networks trained on MegaDepth, seed 20): top-view camera centers in the evaluation frame; whiskers join each reconstructed camera to its ground-truth position. The label-free madweight arm (left) lands on the ground-truth layout (translation median 0.06, rotation 1.0°); the supervised baseline (right) registers the same number of cameras but collapses them toward a line with large systematic displacements (median 2.55, rotation 16.5°) — on a scene with almost no outliers to reject.

5.2 The complementarity finding

Fig. 1 shows Table 2 as a picture, and one pattern organizes it. Between supervision regimes, the self-supervised wins concentrate exactly where a label-free design should help: on clean OOD data, where a classifier trained on contaminated MegaDepth misfires (on BlendedMVS, RESfM’s rotation error explodes to 36°), and at extreme contamination, where supervision itself can no longer be manufactured (§5.3). Within the self-supervised family, the removal-vs-weighting ordering is monotone in contamination: weighting is best at 2–3%, the two tie in distribution, and removal dominates from 43% upward (madweight 12.4 vs. weight 12.8, plain remove 8.9, Table 6; 12.4 ± 2.1 vs. ~ 23 at 60%). This mechanism-level flip, the answer to our central question, is stable across schedules and seeds.

Baseline learning-rate robustness, the full mechanism matrix, the MAD/STD/Huber ablation, per-scene selection, and qualitative results: Apps. B–A.

5.3 Why label-free removal wins under high contamination

Two factors explain why the supervised advantage erodes as contamination rises. First, domain shift: the classifier is trained only on in-distribution MegaDepth, and its decision boundary does not transfer to the outlier statistics of out-of-distribution internet collections (a domain-shift effect, not a labeling failure: MegaDepth’s labels derive from its curated reference reconstructions, the trusted-reconstruction condition under which, per Table 3, supervision is sound, and the strong in-distribution result is consistent with this). Even there the head’s classification is mediocre by RESfM’s own ablation (mean outlier recall 72.1% at precision 44.1%; their Table 7), and their Appendix D attributes much of the final accuracy to robust BA — consistent with our pre/post-BA analysis (App. L): a large part of the supervised system’s robustness is BA repair, not classification. Second, and more fundamentally, supervision itself degrades in this regime: RESfM’s labeling (Appendix C of (Khatib et al., 2025)) assumes a COLMAP reconstruction and ground-truth poses per training scene: readily obtained for curated data, but exactly what one lacks on the contaminated collections where robustness is most needed. madweight requires neither: it recomputes the outlier test per-scene from reprojection statistics, so it is unaffected by domain shift and applicable wherever a partial reconstruction exists. This is why the removal family prevails at extreme contamination (§5.4); madweight shares that robustness while requiring no learned threshold at all.

Table 3: The label ceiling: quality of COLMAP-regenerated supervision vs. contamination.

Contam.	n	Cover%	Prec.	Recall	COLMAP t -err
0–5%	6	99.4	0.76	0.82	0.004
5–15%	4	89.6	0.90	0.83	0.014
15–30%	2	71.7	0.83	0.26	1.02
30–45%	3	55.7	0.74	0.39	0.82
45–60%	3	55.1	0.87	0.39	11.0

Table 4: 1DSfM-hard ($\sim 60\%$ outliers, seed 20): translation/rotation and registered fraction.

Method	Trans/Rot (post-BA)	Registered
remove	10.6/12.4	all
madweight	12.2/14.4	all
remove+TTT	15.2/16.8	all
RESfM-scratch	16.2/22.7	all
huberweight	16.9/24.5	all
COLMAP	20.9/–	64%
weight	21.9/25.6	all
stdweight	23.2/27.3	all
hybrid	26.3/28.7	all
weight+TTT	26.4/32.0	all

Measuring the label ceiling. We make the second argument quantitative. On 18 scenes spanning the full contamination axis we simulate the realistic labeling scenario for a new dataset (no ground-truth poses, so labels must come from a from-scratch COLMAP reconstruction of the same tracks with fixed known intrinsics) and score the resulting per-observation labels against the ground-truth-pose reference labels (precision/recall with outlier as the positive class; coverage is the fraction of observations that receive any COLMAP verdict at all, i.e. camera registered and track triangulated; scenes capped at 120 cameras); Table 3 reports the result. Precision stays high throughout, but recall collapses from 0.82 to 0.26–0.39 and coverage halves once contamination exceeds $\sim 15\%$: the corrupted reconstruction absorbs enough outliers that most reproject acceptably under its distorted geometry, and half the observations receive no label at all. A head trained on such labels is actively taught that most true outliers are inliers, precisely where robustness matters. Conversely, below $\sim 15\%$ COLMAP-derived labels are excellent, consistent with RESfM’s strong in-distribution result and delimiting where the supervised recipe is sound.

5.4 Extreme contamination: the 1DSfM-hard set

To probe beyond 1DSfM’s 43%, we evaluate on the five held-out 1DSfM-hard scenes ($\sim 60\%$ measured contamination, 300 cameras each). The learned arms are those of Table 2 (this restates its 1DSfM-hard column alongside rotation), and COLMAP runs incremental mapping on the same tracks with known intrinsics (Table 4). The mechanism-level ordering extrapolates: over seeds 20–24, plain remove attains 12.4 ± 2.1 translation, below the supervised band (19.2 ± 3.4) on every seed pair (medians 10.8 vs. 16.5); madweight at 18.9 ± 4.1 statistically matches the baseline, and every other mechanism (soft, hybrid, and alternative robust statistics) lands at 16.9 or worse at seed 20 (Table 4), and COLMAP degrades severely: its error is computed only over the 64% of cameras it registers (a favorable metric) yet still loses. Errors are large for every method (these scenes are hard), but the trend matches the label-ceiling prediction: the supervised advantage vanishes exactly where supervision can no longer be manufactured. Strikingly, plain remove on the contaminated tracks also beats ESfM* on oracle-cleaned tracks (12.4 ± 2.1 vs. 18.12 ± 4.35 ; 23 of 25 seed pairs): at 60% contamination, even ground-truth labels used for hard removal destroy more than they fix — deleting 60% of observations starves tracks below the multi-view minimum and thins the viewing graph, whereas adaptive removal with soft-weighted survivors preserves the constraint structure. Having perfect labels would not rescue the removal-by-oracle recipe; the

Table 5: In-distribution comparison on the (formerly OOD) domains, ordered by contamination: mean translation over held-out test scenes, mean $\pm$ std over five training seeds per recipe. Supervised labels are GT-derived (best case). Best per column in bold.

Arm	Olsson @0.5%	BlendedMVS @3.1%	1DSfM @43%	1DSfM-hard @60%
madweight	1.97 $\pm$ 0.94	0.093 $\pm$ 0.099	6.37 $\pm$ 0.20	12.1 $\pm$ 6.1
weight	3.05 $\pm$ 1.39	0.114 $\pm$ 0.006	10.65 $\pm$ 2.49	28.1 $\pm$ 5.9
weight+TTT	1.60 $\pm$ 0.81	0.081 $\pm$ 0.046	9.91 $\pm$ 2.39	29.3 $\pm$ 8.4
remove	2.94 $\pm$ 0.27	0.228 $\pm$ 0.080	6.68 $\pm$ 1.44	19.4 $\pm$ 7.7
RESfM (sup., GT labels)	2.03 $\pm$ 1.86	0.260 $\pm$ 0.149	4.97 $\pm$ 2.08	23.4 $\pm$ 7.8
ESfM (no mechanism)	1.45 $\pm$ 1.44	2.36 $\pm$ 3.27	10.20 $\pm$ 2.97	23.14 $\pm$ 5.89

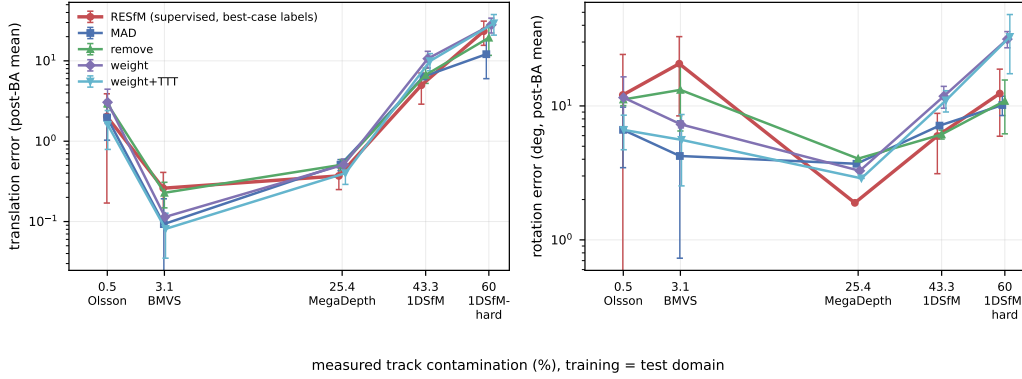


Figure 4: The completed in-distribution contamination curve. Post-BA mean translation (log scale) when training and test share the domain, versus measured track contamination. The supervised recipe uses best-case labels (GT-derived; on MegaDepth, its native COLMAP labels). Each label-free arm is its own line: soft weighting ($\pm$ TTT) leads at low contamination, robust removal (madweight, remove) at high — the complementarity mechanism flip replicated in-distribution. Supervision wins at 25% and narrowly at 43%, is seed-bimodal at 0.5% (mean tied with the label-free arms at more than twice their spread), and loses at 3.1% (dataset-specific dead zone) and 60% (the causal label ceiling): its failures are two distinct modes, not a smooth function of contamination. The rotation panel (right) tells the same story and makes the BlendedMVS dead zone starker still (supervised 20.7 $\pm$ 12.2 $^\circ$ vs. label-free 4–13 $^\circ$). Bands: $\pm$ std over five training seeds (MegaDepth rotation: five-seed means, bands omitted).

label ceiling is thus not the only obstacle at the extreme — how labels are used matters as much as having them.

5.5 Training on the contaminated domains: the label ceiling is causal

The label-ceiling analysis predicts that supervision fails at high contamination because the labels themselves become untrustworthy — not because of domain shift. We test this causally by making the OOD domains in-distribution: for each dataset we split its scenes into train/validation/held-out test and retrain both recipes on the domain itself (identical protocol; supervised labels generated at reference quality, by the same Appendix-C procedure as MegaDepth’s but against each dataset’s real GT poses — the best case supervision could ever obtain). Table 5 reports five-seed bands on the held-out scenes.

Four findings (Fig. 4). (i) At 60%, supervision loses on its own training domain: the supervised head, trained on these scenes with reference-quality labels, is beaten $\sim 2\times$ at band level by madweight (12.1 $\pm$ 6.1 vs. 23.4 $\pm$ 7.8; four of five madweight seeds beat every supervised seed; medians 9.8 vs. 20.6). “Just train the classifier there” does not work: the high-contamination failure is caused by labeling at contamination, not by domain shift.

Sharper still: the supervised head with best-case labels performs identically to no outlier mechanism at all at this contamination (vanilla ESFM trained in-domain: 23.14 ± 5.89 vs. supervised 23.4 ± 7.8) — at 60%, GT-supervised classification adds nothing over doing nothing, while label-free removal halves both. The vanilla row completes a symmetric statement: GT-label supervision matches no mechanism at all at both extremes of the axis — identical at 60% (23.4 vs. 23.1) and overlapping at 0.5% in-domain (2.03 ± 1.86 vs. 1.45 ± 1.44) — earning its keep only in the curated middle band (25–43%). The absolute errors deserve decomposition (Fig. 5): they are not uniformly “large for everyone.” On Roman_Forum, madweight produces usable reconstructions (median 1.6–3.5, rotation 2.3–8.2°, in four of five seeds) where supervision fails outright (medians 9.5–36); on Union_Square the supervised errors look lower but are computed over only 43–55 of 300 registered cameras, versus madweight’s 137–186 — the error-vs-coverage trade-off of §4.2 again, with the label-free arm covering 3–4× more of the scene at moderate error. (ii) The BlendedMVS failure was never domain shift either: supervision trained on BlendedMVS lands at 0.20–0.37 in four of five seeds — indistinguishable from the MegaDepth-trained baseline evaluated OOD (0.35 ± 0.04) — with a single escape seed (0.02), the same bimodality as its clean-ODD behaviour, while label-free weighting is seed-stable at 0.114 ± 0.006 . Class imbalance alone cannot explain this: at 0.5% (Olsson), where positives are rarer still, supervision trained in-domain reaches 0.41–1.08 in three of five seeds — extreme positive scarcity does not preclude learning — whereas BlendedMVS never reaches that level on any seed; the BlendedMVS failure is a dataset-specific dead zone (its rotation pathology is the leading suspect), not a general low-contamination bound. (iii) In-domain training attenuates but does not cure the clean-benchmark unreliability: the recipe that lands at 7.4–9.1 on Olsson MegaDepth-trained improves to 2.03 ± 1.86 trained in-domain — but bimodally, excellent in three seeds (0.41–1.08) and degraded in two (3.9–4.2, rotation spread $12 \pm 12^\circ$). The OOD collapse is thus mostly the MegaDepth-fit head misclassifying on clean scenes (in-domain labels remove the bulk of it), yet the seed-reliability asymmetry survives even here: weight+TTT (1.60 ± 0.81) leads the column, madweight (1.97 ± 0.94) matches the supervised mean at half the spread, and no label-free arm has a failure seed. Supervision’s clean-data advantage is a three-in-five-seeds advantage, not a dependable one. (iv) The supervised advantage inverts between 43% and 60%: at 43% supervision retains an overlapping edge (4.97 ± 2.08 vs. madweight 6.37 ± 0.20 ; the supervised band’s spread is dominated by one strong seed at 1.38), while at 60% it loses $\sim 2\times$ — and the complementarity ordering replicates in-distribution: robust removal (madweight/remove) leads at 43–60% and soft weighting ($\pm$ TTT) at 3% — the mechanism flip is a property of the contamination level, not of the train/test relationship. (A same-protocol Strecha split, one tiny test scene, is an uninformative three-way tie at 0.01.) Read as a curve, supervision with best-case labels wins in distribution at 25% and narrowly at 43%, is seed-bimodal at 0.5% (tied in mean, far wider in spread), and loses at 3.1% and 60%: its failures are not a smooth function of contamination but two distinct modes — a dataset-specific dead zone, and the causal label ceiling at high contamination that no in-domain retraining can cross — with seed-unreliability shadowing it along the whole axis.

5.6 Preliminary: training-pool diversity and the label ceiling

A preliminary experiment (single seed, under a coarser five-milestone decay schedule; cf. §4.2) retrains both methods on progressively diversified pools: MegaDepth-27 $\rightarrow$ +ETH3D (39 scenes, added data $\sim 8\%$ contaminated) $\rightarrow$ +VGG/T&T (52 scenes, adding a $\sim 16\%$ -contaminated mid-band). The supervised baseline’s clean-ODD accuracy degrades monotonically with diversity (Strecha $0.006 \rightarrow 0.77 \rightarrow 2.83$); in distribution the same diversification is nearly free (MegaDepth $0.33 \rightarrow 0.32 \rightarrow 0.44$, the first step within noise), so the collapse is specific to clean OOD generalization, not a general capacity cost. Meanwhile the clean addition yielded no high-contamination gain and only the contaminated mid-band coincided with one (1DSfM 11.0/11.5/8.8; differences within single-seed noise on this benchmark, so consistent with, though not established by, these runs). If it holds, the implication is pointed: the data that helps is precisely the data whose COLMAP labels degrade (cf. the label ceiling), making supervised diversity scaling self-limiting.

Roman Forum (60% contamination), both trained in-domain: BA-aligned camera centers, top view

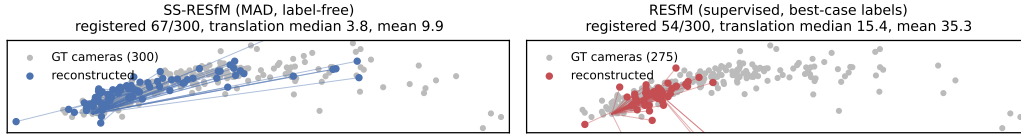


Figure 5: Qualitative view of the 60% label ceiling (Roman Forum, seed 20, both recipes trained in-domain, supervised with GT-derived labels): BA-aligned camera centers in top view; whiskers join each reconstructed camera to its ground-truth position. The label-free madweight arm (left) registers more cameras (67 vs. 54 of 300) that track the ground-truth layout at a usable median error (3.8 vs. 15.4), while the supervised reconstruction (right) collapses toward a compressed cluster with large systematic displacements — supervision fails here with the best labels it could ever have.

A four-cell control (single seed each, canonical schedule) sharpens the architecture question: on the Olsson clean benchmark, the supervised collapse persists for a deep network on the MegaDepth-only pool (8.58) and for a shallow network on the diversified 52-scene pool (9.30), yet vanishes for deep networks on the diversified pool (2.26 plain; 2.83 with Layer-Norm/residual — so normalization is not the ingredient). Neither capacity nor data diversity alone rescues the supervised recipe; only their combination does. The label-free arms reach the same level (2.9–3.6) from the shallow, single-pool configuration — the cheapest cell of the grid — which reframes the comparison: self-supervision buys at 1×3 capacity on 27 scenes what supervision needs both depth and a curated multi-dataset pool to approach.

A two-stage control sharpens what kind of failure the collapse is. Retraining the supervised recipe from scratch on tracks pre-cleaned by its own best outlier head (with label-free reprojection-based checkpoint selection) does nothing at the full 20k+20k budget — all five seeds collapse (8.3–12.0) — but a compute-matched 10k+10k version escapes in three of five seeds (1.69/2.01/2.29, matching the best cells of the grid) while the other two collapse (7.8/12.1), with the same bimodality on Strecha (0.01–0.2 vs. 1.9/2.4). The good basin is therefore reachable from the shallow single-pool configuration — the collapse is basin selection, not a capacity or data impossibility — but no supervised-lineage recipe we tested reaches it reliably: across seeds, clean-OOD reliability is delivered only by the label-free arms and by depth $\times$ diversity. Nor do the two remedies compose: applying the same two-stage cleaning on top of the depth $\times$ diversity configuration destroys its fix (Olsson 9.94 vs. 2.83 one-stage) — pre-cleaning strips the very training signal the diverse pool provides — while incidentally producing the best BlendedMVS result in the study (0.10). Data surgery and data diversity are substitute-seeking interventions that interfere, not stack. The label-free arms diverge by mechanism, and the canonical-schedule three-seed replication (authors’ configuration, 52 scenes) now separates the robust from the fragile: madweight’s clean-OOD diversity-robustness is seed-stable (BlendedMVS 0.12 ± 0.01 vs. the baseline’s 0.35 ± 0.04), whereas its high-contamination gain is real only in the best case: 1DSfM 5.4/11.2/12.8 across seeds (9.8 ± 3.9), overlapping the single-pool band (11.5 ± 2.3). Diversification can yield large gains at 43% but does not do so reliably, and yields none at 60% (20.1 ± 1.2); soft weighting collapses on clean OOD like the baseline at the retired schedule (2.9–4.0) and holds only BlendedMVS under the canonical one.

6 Limitations

(1) No single deployable configuration wins all datasets; selecting the best mechanism per dataset (Table 2) requires knowing the contamination level. We present a study of the

phenomenon, not a turnkey adaptive method; an unsupervised selector is future work. Relatedly, the basin-dependent character of the clean-OOD failures (§5.5; the reliability appendix) suggests a natural remedy we do not pursue here: initializing or anchoring the joint reconstruction with label-free global pose averaging (cf. the concurrent view-graph work of Khatib et al. 2026), which would compose two self-supervised robustness mechanisms at complementary stages of the pipeline — graph-level pose consistency selecting the basin, track-level adaptivity handling observations — toward a fully self-supervised robust SfM pipeline. We leave this composition to future work. (2) Supervised RESfM retains in-distribution MegaDepth (the self-supervised arms come within $\sim 38\%$ at the band level, no closer) and the clean Strecha mean, and RESfM-released remains stronger on MegaDepth, so we claim no SOTA. (3) madweight matches rather than beats supervision at moderate and extreme contamination; plain remove leads every baseline seed pairwise at $\sim 60\%$ and on medians at 43%, but its wins come over a reduced registered-camera set (§4.2), and the magnitude of the supervised deficit at BlendedMVS and high contamination is learning-rate-dependent (App. B) — the Olsson collapse and the mechanism flip are not. (4) OOD tracks are self-constructed: OOD numbers are controlled within our pipeline, not comparable to RESfM’s published OOD numbers. (5) Two symmetric-structure scenes (Ellis Island, Tower of London) fail for all methods; these are the cases where excess removal can disconnect the viewing graph (Khatib et al., 2025).

7 Conclusion

SS-RESfM makes deep SfM outlier handling entirely self-supervised while staying within $\sim 38\%$ of the supervised baseline in distribution, winning two of three clean-OOD splits — reliably, on every training seed — and beating it at high contamination on every seed pair, where both beat COLMAP and supervision itself can no longer be manufactured. The optimal mechanism is contamination-dependent: removal at high contamination, reweighting at low. Robust deep SfM should treat outlier handling as adaptive and self-supervised; code, configurations, and the rebuilt OOD tracks will be released upon publication.

Reproducibility Statement

All results are reproducible from the released materials: code (a documented fork of RESfM’s public repository), every training and evaluation configuration file, the rebuilt track suites for the five OOD datasets and the per-domain train/val/test splits of §5.5, and the scripts that regenerate every table and figure from the saved evaluation outputs (including a verification script that recomputes each printed cell). Training protocol, learning rates, schedules, and checkpoint-selection rules are specified in §4.2; seeds are enumerated per experiment (20–24), and every headline comparison reports five-training-seed bands. Sources of irreducible run-to-run variance (robust-BA solver nondeterminism, TTT) are characterized in App. G so that deviations within those bounds can be recognized as environmental. Dataset provenance and licences: all datasets are public academic benchmarks; our track builder and download manifests are included.

Ethics Statement

This work studies robustness of 3D reconstruction on standard public benchmarks of landmark photographs (MegaDepth, 1DSfM, Strecha, BlendedMVS, Olsson’s dataset) and introduces no new data collection, no personal data processing beyond what those established benchmarks already contain, and no human subjects. Improved robust SfM has broadly beneficial applications (mapping, cultural heritage, robotics); we are not aware of ethical concerns specific to this method beyond those generic to 3D reconstruction of public imagery.

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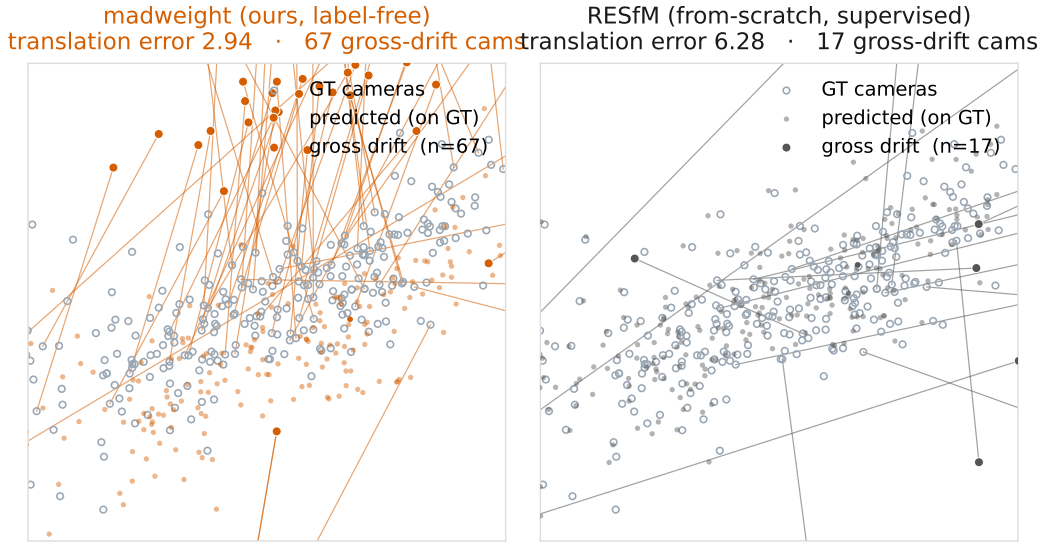


Figure 6: Qualitative 1DSfM (NYC_Library, 32.6% outliers). BA-aligned camera centers vs. GT (open markers); large dots + connectors mark gross-drift cameras. madweight (left, 2.94) has fewer and shorter drifts than RESfM-from-scratch (right, 6.28).

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A Qualitative results

Fig. 6 overlays BA-aligned camera centers on ground truth for 1DSfM NYC_Library. Label-free madweight (2.94) leaves far fewer gross-drift cameras than the supervised from-scratch baseline (6.28), which shows more and larger gross drifts.

B Learning-rate robustness of the baseline

Our protocol gives each method its best learning rate at the canonical 20k budget (10^{-3} for the supervised arms, RESfM’s released value; 10^{-4} for the self-supervised arms). To

Table 6: Full shallow mechanism matrix (RESfM-aligned schedule, seed 20; $\dagger$ mean $\pm$ std over seeds 20–22). Mean / median translation (post-BA); Olsson column: means. Best per column in bold.

Arm	MD	1DSfM	Strecha	BMVS	Olsson
remove	0.60/0.19	$9.1 \pm 2.8 / \mathbf{1.6 \pm 0.2}^\dagger$	3.02/1.68	0.27/0.19	2.81
remove+TTT	0.49/0.13	$\mathbf{8.7 \pm 1.1} / 2.2 \pm 1.1^\dagger$	2.97/1.56	0.19/0.07	4.07
hybrid	0.37/0.20	$16.0 \pm 1.4 / 6.7 \pm 4.4^\dagger$	2.70/1.39	0.10/0.03	2.82
hybrid+TTT	0.39/0.14	$15.4 \pm 1.1 / 10.7 \pm 3.5^\dagger$	3.52/2.65	0.13/ 0.00	3.19
madweight	0.42/0.15	$11.5 \pm 2.3 / 6.0 \pm 1.4^\dagger$	2.50/0.97	0.15/0.004	2.57
weight	0.42/0.18	$15.8 \pm 3.1 / 10.9 \pm 2.1^\dagger$	2.00/0.03	0.10 /0.04	2.65
weight+TTT	0.39/0.14	$17.0 \pm 1.3 / 10.9 \pm 4.1^\dagger$	1.98/ 0.005	0.14/0.01	7.74
RESfM (sup.)	0.34/0.07	11.03/2.93	0.006/0.005	0.38/0.35	8.47

test how much the conclusions depend on this choice we trained the supervised baseline at 10^{-4} as well. In distribution the released value is clearly right (0.344 vs. 0.533), and at 10^{-4} the supervised model is truncation-limited (best checkpoint at the final epoch). Out of distribution, however, the picture splits: the baseline’s Olsson collapse persists at both rates (8.47 vs. 8.54), i.e. the clean-benchmark supervised failure is lr-robust, whereas its Blended-MVS, 1DSfM, and 1DSfM-hard deficits shrink substantially at the lower rate ($0.379 \rightarrow 0.139$, $11.0 \rightarrow 5.9$, $16.2 \rightarrow 12.8$): the under-trained low-rate model overfits MegaDepth less and generalizes better. Two follow-ups complete the grid. A 30k-budget replicate at 10^{-4} plateaus (best validation at epoch 16k, never improving through 30k), and its evaluation sharpens the picture: that checkpoint reaches near-parity in distribution (0.365 vs. the 10^{-3} baseline’s 0.344) while keeping the OOD gains (1DSfM 6.1, 1DSfM-hard 13.9, BlendedMVS 0.18), making it the strongest supervised variant out of distribution (single seed). The Olsson (8.27) and Strecha (1.64) failures persist, and label-free plain removal still leads at extreme contamination (12.4 ± 2.1 vs. 13.9); the OOD advantage of the low rate is a regularization effect, not an under-training artifact. And the self-supervised arms show no symmetric effect: trained at 10^{-3} they are worse on every dataset (madweight MegaDepth 0.75 vs. 0.42, 1DSfM 15.7 vs. 12.4, Olsson 10.1 vs. 2.6). Each method’s declared rate is therefore its empirical best-of-two, and the off-rate OOD gain is specific to supervision, consistent with a COLMAP-label-fitted classifier overfitting in distribution. We scope our claims accordingly: the Olsson and Strecha findings and the mechanism flip itself are lr-robust, while the magnitude of the supervised OOD deficit at BlendedMVS and high contamination depends on the baseline’s learning rate.

C Full mechanism matrix

Table 6 completes the mechanism study (all arms, RESfM-aligned schedule, seed 20). Notably, at 43% the plain-removal family leads every arm including the supervised baseline, and the full three-seed band confirms it: remove+TTT 8.7 ± 1.1 mean (every seed below the baseline’s 11.0), plain remove median 1.59 ± 0.18 (vs. 2.9; means 9.1 ± 2.8 , dominated by the two failure scenes). On clean data the soft arms dominate the removal family, reproducing the complementarity. On outlier-free Olsson the grid flattens (all label-free arms 2.6–4.1 vs. supervised 8.5): the clean-benchmark failure is specific to the supervised head, not the mechanism.

D Which robust statistic? MAD vs. STD vs. Huber

madweight removes by a robust statistic of the reprojection error; we ablate the statistic holding the head reweighting fixed: MAD (median + $\alpha \cdot 1.4826$ MAD, 50% breakdown), a Huber M-estimate of scale (IRLS), and non-robust STD (mean + $\alpha\sigma$), all at $\alpha=2$. Table 7 reports the ablation over seeds 20–22: only MAD’s 1DSfM median is stable; Huber’s col-

Table 7: Removal-statistic ablation (head reweighting fixed). Only MAD’s median is seed-stable.

Statistic	1DSfM mean	1DSfM med. s20/21/22	BMVS
MAD (madweight)	11.5±2.3	4.4 / 7.0 / 6.6	0.15
Huber	11.5 ± 1.6	3.8 / 12.4 / 5.3	0.11
STD	15.2±1.2	13.3 / 9.0 / 12.9	0.15

Table 8: Hyperparameter sensitivity (seed 20, single runs; earlier five-milestone schedule for the percentile variants). Left: pseudo-label percentiles. Right: removal threshold τ on the default checkpoint. Mean translation.

Percentiles	MD (weight)	1DSfM (madweight)	τ	MD	1DSfM
10/90	0.49	13.0	0.4	0.58	8.0
15/85	0.51	15.6	0.5	0.40	10.4
20/80*	0.42	12.4	0.6*	0.39	9.1
25/75	0.60	14.9	0.7	0.38	9.7
30/70	0.44	8.8	0.8	0.43	10.0

lapses on one seed, and non-robust STD is consistently poor, its mean/ σ threshold inflated by the very outliers it should catch. On clean data all three are comparable: little is removed either way, the right behavior without outliers. The 50%-breakdown statistic is the only reliably robust choice, and we use MAD in madweight.

E Sensitivity of the self-supervision hyperparameters

SS-RESfM introduces two hyperparameters: the pseudo-label percentiles (default 20/80) and the removal threshold ($\tau=0.6$, RESfM’s released value). Both were fixed a priori, not tuned per dataset; this appendix reports their sensitivity.

Removal threshold. Sweeping $\tau \in \{0.4, \dots, 0.8\}$ on the default-percentile checkpoint (seed 20): MegaDepth varies only 0.38–0.58 (flat within 0.38–0.43 for $\tau \geq 0.5$), Strecha 3.06–3.22, BlendedMVS 0.20–0.27, and 1DSfM 7.97–10.35 with no consistent optimum; the default $\tau=0.6$ is never far from the per-dataset best. The threshold is a non-issue.

Pseudo-label percentiles. Across a $4\times$ range of percentile choices (Table 8), in-distribution accuracy is flat (MegaDepth 0.42–0.60), and 1DSfM varies by ± 3 –6 — an amount comparable to the training-seed bands of Table 2 (± 1.7 –3.0), with no setting consistently best across mechanisms (these are single-seed cells, so part of the spread is seed noise). The default 20/80 sits mid-range everywhere and was chosen before any evaluation. We conclude the method is not sensitively tuned: neither knob, varied broadly, moves results beyond the seed-level noise this paper quantifies everywhere else.

F Per-scene complementarity and the limits of contamination-based selection

Fig. 7 tests the flip per scene: each scene’s outlier rate vs. the signed advantage of the best removal mechanism over soft weighting. On OOD data, weight wins the low-contamination scenes and removal the high-contamination ones; the crossover is bracketed to the sparsely-sampled ~ 3 –29% band.

Does per-scene contamination help selection? No: within each dataset contamination is too homogeneous to trigger different choices, so a contamination-keyed selector (madweight if scene rate $\geq 10\%$, else weight) collapses to per-dataset selection (Table 9). An oracle choosing the best of five mechanisms per scene from ground-truth error would change the picture: MegaDepth 0.42 $\rightarrow$ 0.116, surpassing every fixed arm, the supervised baseline at 0.344, and even RESfM-released at 0.209, and 1DSfM 12.4 $\rightarrow$ 5.8 vs. the baseline’s 11.0. The performance is thus already latent in the label-free mechanism family; selection, not

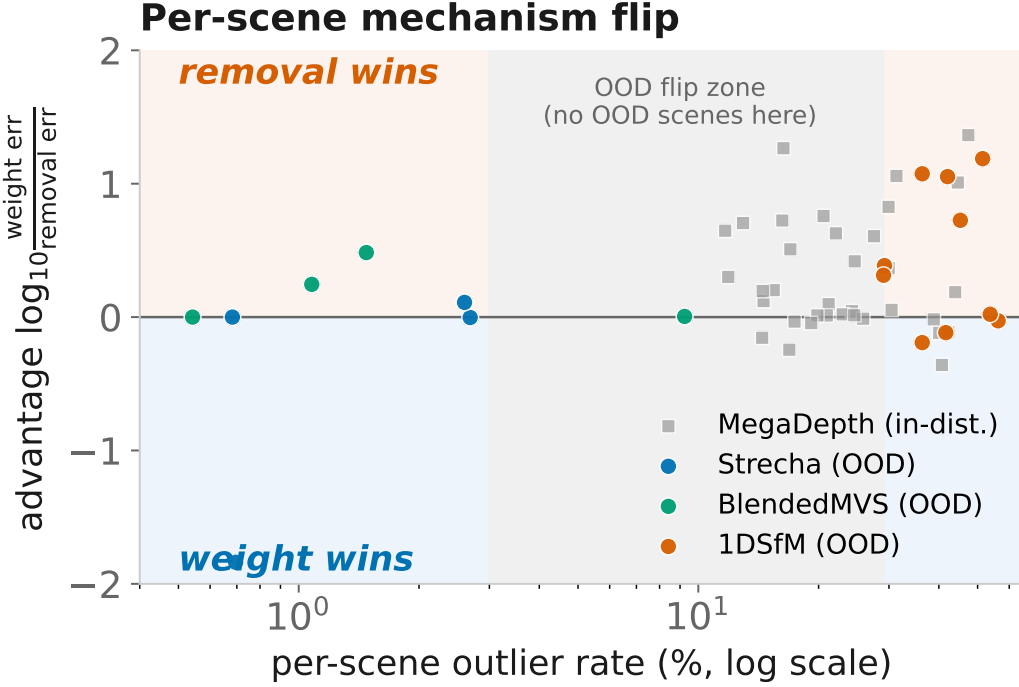


Figure 7: Per-scene complementarity (RESfM-aligned schedule). Each point is one scene: outlier rate (x, log) vs. signed advantage of removal over soft weighting. Weight wins at low contamination, removal at high contamination.

Table 9: Per-scene selection (mean translation; RESfM-aligned schedule). Contamination-keyed selection collapses to per-dataset choice. The oracle uses ground-truth error over five mechanisms: an upper bound, not a deployable method. Self-sel.: run all five, keep the lowest post-BA reprojection error (unsupervised, $5\times$ test-time compute); a secondary observation, not a headline method.

Dataset	weight	madwt	contam	self-sel. ($5\times$)	oracle (GT)
MegaDepth	0.423	0.422	0.422	0.276	<i>0.116</i>
1DSfM	12.79	12.39	12.39	12.83	<i>5.78</i>
Strecha	2.00	2.50	2.00	2.70	<i>1.98</i>
BlendedMVS	0.100	0.146	0.100	0.083	<i>0.083</i>

mechanism design, is the bottleneck. A first unsupervised step exists: running all five mechanisms and keeping the reconstruction with the lowest post-BA reprojection error reaches 0.276 on MegaDepth, past every fixed arm and the supervised baseline (0.344), and matches the oracle on BlendedMVS (Table 9, self-sel.). We report this as a secondary observation rather than a claim: it spends $5\times$ test-time compute, and it inherits reprojection’s retention bias exactly where pruning over-fires (1DSfM 12.8 vs. plain remove’s 8.9). Cheaper selection signals fail informatively. The classification head cannot estimate contamination at all: trained on per-scene percentile pseudo-labels, it is a rank calibrator, not a rate estimator, flagging a near-constant 39–42% of observations on datasets whose true contamination spans 0.5–60% (this also explains the collapse of contamination-keyed selection, and retroactively justifies madweight’s design choice of thresholding reprojection statistics rather than head scores). And pre-fine-tune reprojection magnitudes measure domain shift, not contamination (in-distribution MegaDepth shows the largest raw error); only the post-fine-tune residual tracks the contamination axis monotonically. Realizing the remaining headroom at unit compute is future work.

G Environment sensitivity and reproducibility of post-BA benchmarks

This appendix reproduces the full study behind the protocol’s environment-noise and stack-replication statements (§4.2).

G.1 The reproduction question

Evaluating the RESfM authors’ released checkpoint in our harness on the 36 MegaDepth test scenes yields a post-BA mean translation error of 0.209 vs. the published 0.169 (rotation 1.89° vs. 1.29°). Since our controlled comparisons hinge on baseline fidelity, we traced this discrepancy to its source. The finding is of independent interest: post-BA means on internet-photo benchmarks carry a practical environment-noise floor of $\mathcal{O}(0.05)$, produced by chaotic basin selection in robust bundle adjustment.

Table 10: Released RESfM checkpoint, MegaDepth 36-scene post-BA means, under three conditions, vs. the published number. N_r is the mean number of registered cameras per scene; the published row averages the paper’s per-scene Table 1 values.

Condition	Trans	Rot ($^\circ$)	N_r
Published (paper Table 1)	0.169	1.29	298
Our env (pip/OpenBLAS), 5-seed mean	0.203	1.82	304
Our env, seed 20	0.209	1.89	304
Authors’ exact stack (conda/MKL)	0.217	1.92	304

G.2 Eliminating the controllable factors

Data. Our MegaDepth npz files are byte-identical to the authors’ released archive, including the Group-2 300-camera subsamples.

Randomness. Five evaluation seeds vary the pre-BA network output by ± 0.004 and the post-BA mean by ± 0.01 ; negligible.

Network forward pass. The pre-BA output (translation/rotation $\sim 1.73/11.9^\circ$) is essentially seed- and environment-stable; the divergence therefore enters at the robust-BA stage. (Notably, the paper does not report pre-BA numbers; all its tables are post-robust-BA.)

Software stack. We rebuilt the authors’ exact pinned environment from their released environment.yaml: Python 3.8.20, MKL-backed numpy 1.24.3 (mkl=2023.1.0), pycolmap 3.10.0, pyceres 2.3, lightning 2.3.3. (The sole deviation is the CUDA build of PyTorch, forced by GPU-driver compatibility; the BA stage is CPU-only.) Re-evaluating all 36 scenes in this matched environment gives translation/rotation errors of 0.217/1.92 $^\circ$, statistically indistinguishable from our environment’s 0.209/1.89 $^\circ$, and not the published 0.169/1.29 $^\circ$. Registration is equally stable: all three of our conditions register 304 cameras per scene on average (Table 10), slightly more than the published 298, so the higher mean error is not an artifact of evaluating fewer cameras.

G.3 Mechanism: chaotic basin selection in robust BA

Initialization perturbation is at the 10^{-10} level. We isolated the BLAS layer by running the identical DLT triangulation (the BA initialization) on identical inputs under OpenBLAS and MKL, single-threaded: over 11,131 triangulated points the maximum relative difference is 4.8×10^{-10} (mean 5×10^{-14} ; NaN patterns identical).

Yet outcomes move by $\mathcal{O}(0.1)$. This sub-nanoscale perturbation changes the per-scene post-BA translation by up to 0.167, with 16/36 scenes moving by more than 0.01. Robust BA on ill-conditioned scenes amplifies last-bit differences into different convergence basins.

The published gap has the same signature. Comparing our released-checkpoint evaluation to the paper’s Table 1 per scene: 20/36 scenes agree within 0.05 (8 essentially exactly), 17 are better in our runs, and the entire mean gap is carried by four basin flips (scenes 0046, 0148, 0012, 5016; e.g. 0046: $0.043 \rightarrow 0.892$). Bidirectional, scene-concentrated flips are the signature of basin selection, not of a weaker checkpoint, which would shift errors in one direction rather than improve 17 of 36 scenes.

G.4 Stack replication and TTT run-to-run variance

Beyond the released checkpoint, we re-evaluated both from-scratch main-comparison checkpoints under the authors’ pinned stack (MKL numpy, pycolmap 3.10.0, pyceres 2.3), changing nothing else, and additionally repeated the TTT evaluation within our own stack. The supervised baseline is essentially unchanged across stacks: MegaDepth $0.344 \rightarrow 0.350$ mean, $0.070 \rightarrow 0.088$ median, only 7/36 scenes moving more than 0.05 (max ± 0.37 , bidirectional). This matches the released-checkpoint result above. The SS-RESfM weight+TTT arm, by contrast, varies regardless of stack: three evaluations of the same checkpoint (ours, a same-stack repeat, the authors’ stack) give MegaDepth means 0.316, 0.401, 0.453 (~ 18 – 20 /36 scenes shifting per repeat, up to ± 2.9 , largely the same scenes, medians 0.12–0.19). The attribution is therefore not the software stack but test-time fine-tuning itself: it runs an optimization at evaluation time, so the chaotic basin selection documented above re-enters the result on every run. Consequently the paper reports all results from the single declared pipeline, treats TTT means as carrying $\sim \pm 0.07$ run-to-run variance, and rests its environment- and run-independent claims on the deterministic no-TTT cells (which reproduce exactly on repeat).

G.5 Are seed replicates necessary? Testing seed-additivity

A natural economy argument runs: if the training seed shifted all models by a shared offset (no seed $\times$ model interaction), relative differences would be seed-invariant and a single seed would rank models correctly. We test this directly on the saved per-scene results: for the pairs (supervised – madweight) and (weight – madweight), we form the per-scene, per-seed difference over the five training seeds across all six datasets (98 scene-cells) and ask whether its sign and magnitude are stable.

They are not. At scene level, the sign of the supervised-vs-madweight difference flips across seeds in 67% of scenes (66/98; per-dataset 50–80%), and the median across-seed standard deviation of the difference matches or exceeds the median margin itself (MegaDepth 0.20 vs. 0.10; 1DSfM 4.5 vs. 3.4) — one to two orders of magnitude above the evaluation-seed and environment-noise floors of §G.2, so this is genuine training-seed interaction, not measurement noise. At dataset level, the supervised-vs-madweight verdict itself flips across seeds on four of six datasets: MegaDepth (-0.34 to $+0.10$), 1DSfM (-2.9 to $+3.6$), 1DSfM-hard (-3.1 to $+4.0$), and Olsson ($+5.9$ on four seeds, -0.5 on the escape seed); only Strecha (supervised wins every seed) and BlendedMVS (label-free wins every seed) are seed-invariant — and those are exactly the two cells the paper states as every-seed claims. The within-family pair (weight – madweight) flips only where the two arms are near-tied ($|\text{margin}| \leq 0.3$ on MegaDepth, BlendedMVS, Olsson — flips around zero, i.e. genuine ties), with one true instability at 60% contamination, where soft weighting’s dataset-level margin swings from -2.3 to $+9.7$.

The conclusion is the methodological premise of this paper: seed-additivity fails in this benchmark, a single-seed comparison can reverse the headline verdict on four of six datasets, and any cross-recipe claim must therefore be established at the level of seed bands (or, for the strongest claims, all seed pairings), as done throughout.

Couldn’t one just pick a good seed? A natural rejoinder is that a single-seed comparison is fine if every method reports a well-chosen seed. Four facts close this off. (1) It is not the protocol in use: RESfM fixes one shared seed (20) for itself and for the baselines it retrains, with no selection step — a single lottery draw, and under the interaction above, a lottery whose variance differs by method. (2) Choosing a good seed requires knowing which seed is good, and for the OOD regimes where the failures live, nothing available at

training time identifies it: label-free proxy selectors fail to find the escape seed (App. F), and the authors’ own Accuracy-based checkpoint selection retains the Olsson collapse (8.8). Selecting the seed that avoids a clean-benchmark failure requires evaluating on the clean benchmark itself — test-set selection. (3) The paper’s strongest claims are constructed to survive even adversarially favorable seed choices: at 60%, plain remove’s worst seed beats the supervised baseline’s best (15.8 vs. 16.2), so no seed policy on either side reverses it; the one claim a lucky seed softens (the Olsson margin, where the escape seed ties the label-free band) is stated as a reliability claim — collapse in 4/5 seeds — for precisely this reason. (4) Empirically, the fixed seed was not a lucky one anyway: at seed 20 the baseline is mid-band on MegaDepth and a collapse seed on Olsson, and the released checkpoint (0.209) is itself a favorable draw that the authors’ own training code does not reproduce (0.34–0.39 on our tracks) — consistent with published single-seed numbers being draws from a wide distribution rather than representative values.

Classical baselines and run-to-run variance. The same replicate protocol applied to the classical rows sharpens the tiering: GLOMAP is deterministic in practice (five independent runs per dataset; run-to-run std ≤ 0.06 translation everywhere, e.g. MegaDepth 3.33 ± 0.03 , 1DSfM 27.21 ± 0.06), so its single-run rows are representative; incremental COLMAP is not (20.4 ± 1.1 on 1DSfM-hard over five runs, with per-scene registration swinging by up to 67 cameras) — RANSAC-driven incremental reconstruction inherits real run-to-run variance. The volatility hierarchy is thus: classical global (none) < classical incremental (moderate) < learned recipes (dominant, and concentrated in the supervised arm). A full COLMAP grid (five runs per scene on every dataset) adds nuance: its run-to-run variance is itself contamination-dependent — bit-stable on the clean sets (Olsson 0.20 ± 0.00 , Strecha 0.03, BlendedMVS 0.01, where it is the strongest method in the table) and genuinely variable only at 60% (20.4 ± 1.1). Incremental reconstruction is excellent and reliable exactly where outliers are absent, and fails where they dominate — the classical mirror of the complementarity finding. (COLMAP errors are computed over its registered cameras only, a favorable convention; its two largest 1DSfM scenes are excluded pending completion.)

Even oracle-clean tracks do not confer reliability. ESfM* — the architecture trained on ground-truth-cleaned tracks, with no outlier mechanism in the loop at all — is itself seed-bimodal on BlendedMVS: five training seeds give [0.00, 0.33, 4.93, 5.64, 8.08] (3.79 ± 3.51). The clean-OD reliability failure therefore lives in the geometric trunk’s training dynamics, not in outlier handling or label quality, and no amount of label perfection removes it — only the adaptive label-free arms happen to escape it in our grid.

G.6 Implications for reading the paper’s numbers

(1) The published number and ours are draws from a distribution over environments; no software matching can force a particular draw. (2) Differences of $\lesssim 0.05$ in post-BA translation means on these benchmarks are within environment noise and should not be read as method differences, by any paper. (3) Consequently, meaningful comparisons must be environment-controlled: every number in the main paper (all arms, RESfM-from-scratch, and the released checkpoint) is produced in a single environment, and the out-of-distribution margins ($2\times$ and larger) sit far above this noise floor.

H Per-scene results and configuration

H.1 Per-scene results

Tables 11 and 12 report the per-scene post-BA translation and rotation errors and registered-camera counts underlying the main paper’s dataset-level numbers (seed 20, the declared pipeline): the 36 MegaDepth test scenes and the ten 1DSfM scenes, for the three self-supervised arms (madweight, weight, weight+TTT) and the supervised RESfM baseline trained from scratch under the identical recipe. The per-scene view makes two aggregate statements concrete: on 1DSfM no single arm dominates, the best mechanism flips scene by scene (with Ellis_Island and Tower_of_London hard for every arm), consistent with

Table 11: Per-scene MegaDepth results (layout follows RESfM’s Table 1). For each scene we show the number of input images (N_c) and the fraction of outlier observations; for each model, the number of registered cameras (N_r) and mean rotation (e_R , degrees) and translation (e_t) errors, post-BA, seed 20. Above the middle rule are Group 1 scenes (< 1000 images); below are Group 2 scenes (> 1000 images, subsampled to 300 for testing), each group ordered by outlier fraction. Winning results per scene are marked in bold and underlined. weight+TTT values are single draws of a run-to-run nondeterministic evaluation (Sec. G.4).

Scene	N_c	%out	madweight			weight			weight+TTT			RESfM		
			N_r	e_R	e_t	N_r	e_R	e_t	N_r	e_R	e_t	N_r	e_R	e_t
0238	522	45%	437	19.02	1.41	137	8.23	0.62	141	8.37	0.63	355	<u>1.40</u>	<u>0.12</u>
0060	528	42%	469	0.29	0.03	476	0.20	0.02	452	4.13	0.37	494	<u>0.04</u>	<u>0.01</u>
0197	870	41%	470	7.22	0.89	341	3.54	0.38	422	6.94	1.11	739	<u>0.97</u>	<u>0.11</u>
0094	763	40%	493	2.72	0.45	505	1.44	<u>0.16</u>	432	<u>1.34</u>	0.17	572	5.33	0.88
0265	571	39%	457	17.62	3.26	477	5.78	2.34	547	<u>0.60</u>	<u>0.19</u>	424	16.27	3.09
0083	635	31%	342	4.34	0.23	511	17.70	1.56	317	3.20	0.20	582	<u>1.40</u>	<u>0.10</u>
0076	558	30%	423	1.40	0.38	519	0.61	0.23	454	0.74	0.22	499	<u>0.25</u>	<u>0.02</u>
0185	368	30%	347	0.21	0.01	335	0.24	0.04	321	0.88	0.09	352	<u>0.07</u>	<u>0.01</u>
0048	512	24%	475	1.93	0.14	467	<u>0.35</u>	<u>0.02</u>	450	0.43	0.03	489	1.85	0.11
0024	356	23%	299	1.27	0.18	291	<u>1.13</u>	<u>0.17</u>	289	4.78	1.13	334	3.67	1.09
0223	214	17%	205	2.66	0.24	198	1.69	0.23	199	3.06	0.30	204	<u>0.45</u>	<u>0.08</u>
5016	28	17%	23	0.27	0.04	25	0.08	0.02	28	<u>0.07</u>	0.02	28	0.08	<u>0.01</u>
0046	440	15%	422	<u>2.88</u>	<u>0.22</u>	421	4.11	0.30	419	3.88	0.25	410	14.64	0.93
0099	299	47%	224	<u>0.81</u>	0.15	216	2.77	3.52	212	1.58	<u>0.13</u>	252	2.18	0.39
1001	285	44%	280	1.58	1.22	278	1.49	0.73	279	<u>1.22</u>	<u>0.49</u>	273	1.23	0.50
0231	296	42%	274	1.63	0.15	257	0.64	0.07	264	1.49	0.24	265	<u>0.17</u>	<u>0.02</u>
0411	299	30%	281	0.22	0.03	245	2.99	0.32	250	4.12	0.46	275	<u>0.11</u>	<u>0.02</u>
0377	295	28%	200	0.62	0.05	222	2.76	0.26	243	<u>0.56</u>	<u>0.03</u>	223	0.82	0.06
0102	299	26%	289	0.15	0.02	289	<u>0.07</u>	<u>0.01</u>	291	0.08	0.01	287	0.10	0.02
0147	298	25%	261	31.69	2.99	235	15.28	1.21	225	15.27	<u>1.20</u>	247	<u>4.61</u>	3.22
0148	287	25%	220	2.60	0.33	201	2.11	0.11	196	<u>0.62</u>	0.04	202	0.94	<u>0.04</u>
0446	298	22%	292	<u>0.40</u>	<u>0.03</u>	285	1.20	0.19	292	1.04	0.19	293	0.71	0.05
0022	297	21%	275	0.14	0.02	272	0.22	0.03	267	0.33	0.04	276	<u>0.11</u>	<u>0.02</u>
0327	298	21%	282	1.97	0.09	285	0.22	0.02	284	0.21	0.02	277	<u>0.09</u>	<u>0.01</u>
0015	284	21%	196	2.64	0.30	196	3.87	0.56	211	3.90	0.65	232	<u>1.31</u>	<u>0.30</u>
0455	298	20%	283	<u>0.20</u>	<u>0.01</u>	285	0.21	0.01	288	0.46	1.60	293	0.85	0.09
0496	297	19%	279	<u>0.27</u>	0.04	285	0.50	<u>0.04</u>	287	1.28	0.09	283	0.46	0.07
1589	299	17%	265	0.96	0.06	262	0.28	0.05	296	1.91	0.30	291	<u>0.06</u>	<u>0.01</u>
0012	299	16%	281	<u>0.23</u>	<u>0.02</u>	247	9.51	0.70	262	2.76	0.22	288	5.53	0.34
0104	284	16%	189	0.32	0.03	193	2.28	0.18	194	1.44	0.07	199	<u>0.30</u>	<u>0.02</u>
0019	299	15%	232	3.22	0.17	253	6.55	0.28	264	3.24	0.13	252	<u>0.05</u>	<u>0.01</u>
0063	293	14%	258	0.63	0.08	268	0.72	0.07	267	0.65	<u>0.06</u>	266	<u>0.44</u>	0.07
0130	285	14%	250	10.17	1.60	184	1.87	0.16	185	<u>0.33</u>	<u>0.01</u>	195	1.03	0.07
0080	284	13%	132	1.10	<u>0.05</u>	148	2.48	0.20	147	0.70	0.07	141	<u>0.66</u>	0.10
0240	298	12%	275	<u>2.23</u>	<u>0.25</u>	275	4.64	0.34	272	5.32	0.46	282	3.02	0.35
0007	290	12%	106	<u>0.07</u>	<u>0.01</u>	269	1.86	0.09	261	3.28	0.14	104	0.99	0.04

the per-scene complementarity analysis in the main paper; on MegaDepth the four arms are close on most scenes and the dataset-level gap is dominated by a handful of high-error scenes. Per scene, the catastrophic scenes (Ellis_Island, Tower_of_London) fail in both metrics, though rotation does not always track translation (weight’s Vienna_Cathedral failure is translation-specific: 22.71/0.46), and N_r stays in the 58–92% range of each scene’s images on 1DSfM with no arm systematically dropping scenes. The dataset-level rotation and registration record accompanies the main comparison (Table 2). Rotation broadly agrees with translation at the dataset level: the supervised BlendedMVS failure is even larger in rotation (36°) and the high-contamination ordering is preserved, though the best arm can differ per metric (on MegaDepth, weight+TTT leads translation while RESfM leads rotation). Registration is contamination-dependent: the supervised baseline registers the most cameras in distribution (85% vs. 78–80% on MegaDepth) and on clean Strecha, while

Table 12: Per-scene 1DSfM results (layout follows RESfM’s Table 1). Columns as in Table 11; seed 20, scenes ordered by outlier fraction. Winning results per scene are marked in bold and underlined.

Scene	N_c	%out	madweight			weight			weight+TTT			RESfM		
			N_r	e_R	e_t	N_r	e_R	e_t	N_r	e_R	e_t	N_r	e_R	e_t
Ellis_Island	227	56%	180	<u>17.16</u>	17.59	171	17.37	<u>15.90</u>	177	17.74	17.11	192	18.15	16.02
NYC_Library	332	54%	282	2.32	2.94	261	<u>1.16</u>	<u>1.19</u>	279	10.30	3.39	285	12.99	6.28
Vienna_Cathedral	836	51%	537	7.42	7.30	484	<u>0.46</u>	22.71	651	26.80	33.57	492	1.09	<u>2.29</u>
Notre_Dame	553	48%	507	3.33	0.68	491	1.00	0.34	485	<u>0.83</u>	<u>0.33</u>	493	1.08	0.49
Yorkminster	437	45%	299	8.05	<u>5.79</u>	371	14.95	30.87	380	12.20	10.79	309	<u>6.88</u>	8.77
Piazza_del_Popolo	338	42%	218	<u>0.43</u>	<u>0.94</u>	265	19.54	10.66	254	20.46	17.05	255	3.15	2.49
Tower_of_London	472	42%	317	35.61	76.34	337	23.30	<u>28.29</u>	314	<u>17.91</u>	72.61	299	49.66	67.95
Montreal_Notre_Dame	450	36%	311	0.59	0.94	325	<u>0.55</u>	11.35	320	0.81	1.24	400	0.59	<u>0.93</u>
Alamo	577	29%	467	2.45	1.77	507	2.74	2.18	497	2.49	<u>1.67</u>	502	<u>2.42</u>	1.73
Madrid_Metropolis	341	29%	249	5.72	9.63	301	2.84	4.37	295	2.35	3.52	287	<u>1.82</u>	<u>3.38</u>

Table 13: Training and evaluation configuration. All arms in the main paper use these settings; the only per-method difference is the base learning rate.

Setting	Value
Training (both methods)	
Epochs	20,000 (one epoch = one pass over all scenes)
Batch	4 scenes
LR schedule	milestones 10k, 11k, ..., 20k, $\gamma = 0.9$
Base LR	10^{-4} (SS-RESfM) / 10^{-3} (RESfM)
Loss weights	$\alpha_{\text{reproj}} = 1$, $\beta_{\text{cls}} = 1$
Infinity-point margin	10^{-4}
Checkpoint selection	best validation error, evaluated every 500 epochs
Seed	20 (21–22 / 21–24 for the seed studies)
Adaptive pseudo-labels (SS-RESfM only)	
Inlier / outlier percentile	20 / 80 (of per-observation reproj. error)
Threshold separation	scale-aware floor, $0.25 \times$ median error
Warm-up	10 epochs before pseudo-labeling starts
Evaluation	
Removal threshold τ	0.6 (0.8 for low-contamination single-camera sets, following RESfM)
Test-time fine-tuning	1,001 iterations, LR 5×10^{-3} , evaluated every 250 iterations
Bundle adjustment	pyceres/pycolmap robust BA, as RESfM

under high contamination the label-free arms retain more cameras than supervision (38–43% vs. 37% on 1DSfM-hard) at lower error (madweight).

I Comparison with ESfM in both of its regimes

ESfM is the no-outlier-mechanism ancestor of this model family and operates in two regimes: multi-scene training and its original per-scene optimization. We compare in both.

Multi-scene (same contaminated tracks). Vanilla ESfM — the ESfM architecture inside the RESfM codebase, so trainer, tracks, and BA are identical to every other arm — trained under the unified protocol (reference row in the main table; five-seed bands) reaches 0.71 ± 0.11 on MegaDepth, 18.42 ± 0.75 on 1DSfM, 21.76 ± 0.80 at 60%, 2.03 ± 0.14 on Strecha, 6.45 ± 3.75 on BlendedMVS, and 5.91 ± 3.32 on Olsson. Three readings. First, without any outlier mechanism, contaminated training data degrades every contaminated benchmark relative to all mechanism-equipped arms — the premise of RESfM and of this paper. Second, its apparent BlendedMVS competitiveness at seed 20 (0.23) dissolves at five seeds ($6.45 \pm$

Table 14: Cross-generalization grid: rows = training domain, columns = test dataset; each cell madweight / supervised (mean $\pm$ std over training seeds). Diagonal (*) = held-out in-distribution test (Table 5); “leak” = training scenes overlap the target.

Trained on $\downarrow$	MegaDepth	Olsson	Strecha	BMVS	1DSfM	1DSfM-hard
Olsson	0.52 / 0.62	*1.97 / 2.03	3.05 / 0.72	6.97 / 6.70	11.84 / 12.04	15.56 / 20.17
Strecha (2 seeds)	0.53 / 0.54	2.69 / 8.62	*0.01 / 0.01	7.29 / 8.08	13.94 / 10.22	17.06 / 18.97
BMVS	0.63 / 0.51	2.73 / 6.95	2.56 / 1.44	*0.09 / 0.26	11.50 / 10.45	19.70 / 18.59
1DSfM-easy	0.53 / 0.56	3.27 / 7.20	2.94 / 1.92	6.48 / 5.81	*6.37 / 4.97	18.31 / 20.12
1DSfM-hard	0.47 / 0.83	2.93 / 7.78	3.02 / 1.94	3.65 / 5.68	13.21 / 8.49	*12.1 / 23.4
1DSfM-merged	0.52 / 0.59	3.34 / 4.96	2.98 / 2.37	7.26 / 6.69	*6.71 / 5.05	leak

3.75, bimodal) — one more single-seed verdict reversed by banding (cf. the seed-additivity analysis) — and its Olsson behaviour is likewise bimodal (collapse in three of five seeds), so the clean-OOD failure mode predates the supervised head and lives in the trunk. Third, on Strecha the band (2.03 ± 0.14) statistically ties the best label-free soft arms (1.97–2.00), confirming at band level that at the clean extreme the correct mechanism approaches none.

Single-scene (ESFM’s original per-scene optimization, Olsson). Here the primary ESFM row runs in the authors’ official ESFM codebase; the RESfM-codebase reimplementation is included as a cross-check and agrees to within noise (0.19 vs. 0.22°), ruling out implementation gaps between the two regimes. On 36 calibrated Olsson scenes (3 runs each, matching ESFM’s protocol), post-BA median-over-scenes rotation/translation:

Method (per-scene optimization)	Rot ($^\circ$)	Trans
ESFM (official code)	0.19	0.057
ESFM (RESfM codebase)	0.22	0.088
ESFM + sequential init (10 scenes)	0.10	0.032
ESFM + MAD statistic	0.68	0.123
self-supervised arm (base)	8.03	0.748
self-supervised arm (30/70 percentiles)	2.81	0.434
self-supervised arm + TTT	1.22	0.324

Vanilla ESFM wins this regime outright, and every outlier mechanism — statistical or learned — costs accuracy. This is not a failure of the mechanisms but the low-contamination end of the complementarity finding replicated in a second regime: at 0.5% contamination with per-scene optimization there is nothing to reject, and any removal or reweighting machinery only discards signal. The practical reading matches the main paper: mechanism choice must adapt to contamination, and “none” is the correct mechanism at the clean extreme of the axis.

J Cross-generalization: every training domain, every test domain

We train both recipes on each of the six domains (in-distribution protocol of §5.5: GT-derived labels for the supervised arm) and evaluate every model on all datasets ($\sim 5,400$ evaluations; five training seeds per cell, strechaid two). Cells where a pool’s training scenes overlap the target are excluded. Table 14 reports madweight / supervised post-BA translation means.

Three findings generalize beyond MegaDepth training. (i) The clean-benchmark collapse is training-domain-invariant: the supervised recipe degrades on Olsson OOD from every training domain (4.96–8.62 across five pools, seed-volatile up to ± 4.25), while madweight lands in a tight 2.69–3.34 band from every pool — the reliability asymmetry is a property of the recipes, not of MegaDepth. (ii) The Strecha exception is likewise invariant: supervision wins Strecha from every training domain (0.72–2.37 vs. 2.56–3.05). (iii) At the extreme (60% target), madweight from any domain beats supervision from any domain (best supervised cell 18.59 vs. madweight 15.56–19.70, with the in-distribution madweight at 12.1) —

Table 15: Per-scene label-degradation results. For each scene: contamination (fraction of track observations that are GT outliers), COLMAP-label coverage (fraction of observations receiving a label), and precision/recall of the COLMAP-derived outlier labels against GT, sorted by contamination. Above $\sim 15\%$ contamination, label recall never exceeds 0.50: the labels miss at least half of the true outliers.

Dataset	Scene	Contam. (%)	Coverage	Prec.	Recall
BlendedMVS	58c4bb4f4a...	0.5	1.00	0.86	0.94
Strecha	Herz-Jesu-P8	0.7	1.00	0.51	0.66
Strecha	fountain-P11	0.7	1.00	0.69	0.76
Strecha	Herz-Jesu-P25	2.6	0.99	0.72	0.82
Strecha	entry-P10	2.7	0.99	0.88	0.88
BlendedMVS	5be883a4f9...	4.0	0.98	0.90	0.84
BlendedMVS	5b6e716d67...	6.1	0.92	0.89	0.92
BlendedMVS	5bfd0f32ec...	7.1	0.97	0.97	0.97
BlendedMVS	5b950c7160...	9.2	0.96	0.92	0.94
MegaDepth	0007	11.2	0.73	0.84	0.47
1DSfM	Alamo	26.6	0.76	0.81	0.25
1DSfM	Madrid_Metropolis	28.8	0.67	0.85	0.28
MegaDepth	0041	31.4	0.49	0.80	0.42
MegaDepth	0060	39.0	0.42	0.53	0.45
1DSfM	Piazza_del_Popolo	41.9	0.76	0.90	0.30
MegaDepth	0047	54.9	0.37	0.83	0.50
1DSfM	Ellis_Island	57.9	0.79	0.89	0.37
1DSfM-hard	Gendarmenmarkt	59.8	0.50	0.89	0.29

and, remarkably, MegaDepth as a test set is nearly training-domain-independent (all cells 0.47–0.83), suggesting its curated test scenes are easy for any reasonably trained model of this family. One counterexample sharpens the picture: supervised trained on 1DSfM-hard generalizes to standard 1DSfM better than anything else (8.49), consistent with supervision learning transferable contamination statistics when its labels are sound on a related domain.

K Per-scene label degradation

Table 15 gives the per-scene data behind the main paper’s label-ceiling analysis: for each of the 18 scenes we measure the contamination level, the fraction of track observations that receive a COLMAP-derived label at all (coverage), and the precision and recall of those labels against ground truth. The aggregate trend in the main paper is visible scene by scene: below $\sim 10\%$ contamination coverage is 0.92–1.00 and recall is 0.66–0.97 (mostly above 0.8), while above $\sim 15\%$ contamination recall never exceeds 0.50 on any scene (0.25–0.50) and coverage falls as low as 0.37: the supervision signal degrades exactly where outlier handling matters most.

L Pre- versus post-BA seed variability

Table 16 reports, for every arm and dataset, the five-seed band of the network output (pre-BA) next to the band after robust bundle adjustment. The behaviour is what one expects of a non-linear optimiser with multiple basins, and it is worth recording because two statements in the main text depend on it.

A direct control separates the two sources. Re-evaluating the authors’ fixed released weights under five evaluation seeds gives 0.203 ± 0.008 on MegaDepth (pre-BA 1.730 ± 0.005), 10.71 ± 0.27 on 1DSfM, and identical values to two decimals on Strecha (0.20) and BlendedMVS (0.33): with the network held constant, bundle adjustment is close to deterministic, and exactly deterministic on the small clean sets. The large bands below therefore reflect genuine differences between independently trained networks, amplified by BA, and not evaluation noise. A full paired-seed replication confirms this at scale: re-running all 2,352 evaluations with the evaluation seed set equal to each cell’s training seed (rather than fixed at 20)

reproduces every five-seed band within a fraction of its std — e.g. the baseline row moves from 0.37 ± 0.12 to 0.38 ± 0.14 on MegaDepth and is unchanged to two decimals on Strecha, BlendedMVS, and Olsson — with the largest shifts in the TTT cells (~ 0.13), consistent with their run-to-run nondeterminism. The choice of evaluation seed is immaterial to every conclusion in this paper.

First, on the contaminated datasets nearly all seed variability is created by bundle adjustment, not by training: pre-BA bands are ± 0.02 – 0.13 for the self-supervised arms and ± 0.05 – 0.43 for the baseline (relative spreads of $\sim 0.5\%$ and 1 – 3% respectively), while the same cells after BA carry ± 1.7 – 4.1 . The networks are highly reproducible across seeds; most of the pipeline’s variance enters at the BA stage, which lowers the error on 234 of 242 baseline scene evaluations we examined while widening the relative seed band by an order of magnitude — BA compresses the mean and amplifies the spread. This is the seed-level counterpart of the environment-level chaotic basin selection documented in App. G, now measured over 36 independent cells.

Second, the removal arms do not produce better raw geometry than supervision — they produce geometry that robust BA can repair. Pre-BA the supervised baseline leads everywhere (MegaDepth 1.76 vs. ~ 3.03 ; 1DSfM 22.1 vs. 25.3 – 26.6), yet plain removal improves by $2.9\times$ through BA at 60% contamination ($36.3 \rightarrow 12.4$) against the baseline’s $1.9\times$ ($35.8 \rightarrow 19.2$), and by $5.0\times$ on Olsson ($17.1 \rightarrow 3.4$) against $1.5\times$ ($11.0 \rightarrow 7.4$). Feeding BA a cleaner observation set, rather than a more accurate initialisation, is the mechanism behind the high-contamination result.

A third observation concerns the baseline specifically, and is the one exception to the first: its clean-OOD instability is already present before BA (Strecha 0.68 ± 0.78 , Olsson 11.01 ± 2.27 , versus ± 0.18 – 0.24 and ± 0.19 – 0.47 for the self-supervised arms), locating that failure in supervised training rather than in bundle adjustment or evaluation. Per scene the failure is bimodal and seed-contingent rather than noisy: entry-P10 costs 0.04 – 0.06 on three seeds and $4.0/7.0$ on the other two; dtu106 sits at 250 – 260 on four seeds and 53 on one. Five authors-faithful retrains (seeds 20–24, the authors’ validation metric and margin) reproduce the scratch baseline’s band on every dataset — including the one-bad-Strecha-seed pattern and the four-of-five Olsson collapse (8.66 – 9.02 with a single 2.32 escape, mirroring the scratch 2.69) — so these failure modes belong to the supervised recipe, not to our retraining of it. The same signature appears in-distribution at smaller amplitude — individual MegaDepth scenes swing 3 – $4\times$ across seeds (0060: 1.0 – 3.3 ; 0231: 0.5 – 2.2) while opposite-signed swings cancel in the dataset mean, and the baseline’s median per-scene std is about twice the arms’ (0.11 vs. 0.04 – 0.05). Our reading is structural: the supervised head fits fixed labels generated at $\sim 25\%$ contamination, many decision boundaries achieve near-identical training loss, and its behaviour on scenes far from that regime — near-outlier-free ones in particular — is underdetermined and decided by the draw of the seed. The self-supervised arms, trained against their own reprojection-error percentiles, converge to nearly the same solution from every seed (per-scene stds as low as 0.01). Stability is not accuracy — the arms remain worse pre-BA on clean data — but predictability under retraining is a property the label-free objective buys that labels do not. The single-run reference rows in the lower block sharpen the same point: every supervised variant, including the authors’ released weights, leaves BA with an Olsson error of 8.1 – 9.1 despite entering it at 11.9 – 12.1 — a repair factor of only $\sim 1.4\times$, against 4.7 – $5.5\times$ for the label-free arms from a worse starting point (13.4 – $17.1 \rightarrow 2.9$ – 3.6). On the clean benchmark the supervised head does not give bundle adjustment a set of observations it can fix.

Table 16: Pre-BA $\rightarrow$ post-BA translation error, mean $\pm$ std over five training seeds (seeds 20–24). Pre-BA bands are one to two orders of magnitude tighter than post-BA bands, except the supervised baseline’s clean-OOD cells (Strecha, Olsson), whose variance predates BA (see text). [§]For the released checkpoint the MegaDepth entry is a band over five evaluation seeds on the fixed weights (± 0.008), for comparison with the training-seed bands above: re-seeding the evaluation moves the result by two orders of magnitude less than retraining does.

Arm	MegaDepth	1DSfM	1DSfM-hard	Strecha	BlendedMVS	Olsson
madweight	3.03 $\pm$ 0.02 $\rightarrow$ 0.51 $\pm$ 0.08	26.14 $\pm$ 0.04 $\rightarrow$ 11.04 $\pm$ 1.68	36.33 $\pm$ 0.07 $\rightarrow$ 18.92 $\pm$ 4.05	3.77 $\pm$ 0.24 $\rightarrow$ 2.65 $\pm$ 0.25	0.58 $\pm$ 0.03 $\rightarrow$ 0.14 $\pm$ 0.02	16.38 $\pm$ 0.45 $\rightarrow$ 3.01 $\pm$ 0.34
weight	3.04 $\pm$ 0.03 $\rightarrow$ 0.51 $\pm$ 0.09	26.60 $\pm$ 0.09 $\rightarrow$ 14.35 $\pm$ 2.97	36.26 $\pm$ 0.04 $\rightarrow$ 22.92 $\pm$ 2.79	2.57 $\pm$ 0.19 $\rightarrow$ 1.97 $\pm$ 0.09	0.51 $\pm$ 0.01 $\rightarrow$ 0.14 $\pm$ 0.03	15.86 $\pm$ 0.28 $\rightarrow$ 2.91 $\pm$ 0.33
weight+TTT	3.01 $\pm$ 0.03 $\rightarrow$ 0.40 $\pm$ 0.11	26.63 $\pm$ 0.10 $\rightarrow$ 15.72 $\pm$ 2.10	36.20 $\pm$ 0.12 $\rightarrow$ 22.24 $\pm$ 3.17	2.63 $\pm$ 0.21 $\rightarrow$ 2.00 $\pm$ 0.05	0.50 $\pm$ 0.02 $\rightarrow$ 0.14 $\pm$ 0.02	13.41 $\pm$ 0.19 $\rightarrow$ 8.03 $\pm$ 0.30
remove	3.03 $\pm$ 0.02 $\rightarrow$ 0.51 $\pm$ 0.08	25.34 $\pm$ 0.10 $\rightarrow$ 9.13 $\pm$ 2.17	36.32 $\pm$ 0.09 $\rightarrow$ 12.42 $\pm$ 2.05	4.22 $\pm$ 0.19 $\rightarrow$ 3.20 $\pm$ 0.11	0.62 $\pm$ 0.01 $\rightarrow$ 0.24 $\pm$ 0.06	17.05 $\pm$ 0.39 $\rightarrow$ 3.41 $\pm$ 0.57
remove+TTT	3.04 $\pm$ 0.02 $\rightarrow$ 0.56 $\pm$ 0.07	25.23 $\pm$ 0.13 $\rightarrow$ 8.93 $\pm$ 1.20	36.32 $\pm$ 0.07 $\rightarrow$ 15.15 $\pm$ 0.98	4.31 $\pm$ 0.18 $\rightarrow$ 3.01 $\pm$ 0.07	0.63 $\pm$ 0.02 $\rightarrow$ 0.22 $\pm$ 0.10	17.03 $\pm$ 0.47 $\rightarrow$ 3.63 $\pm$ 0.35
RESM-scratch	1.76 $\pm$ 0.05 $\rightarrow$ 0.37 $\pm$ 0.12	22.13 $\pm$ 0.27 $\rightarrow$ 10.50 $\pm$ 1.45	35.80 $\pm$ 0.43 $\rightarrow$ 19.16 $\pm$ 3.42	0.68 $\pm$ 0.78 $\rightarrow$ 0.39 $\pm$ 0.72	0.45 $\pm$ 0.01 $\rightarrow$ 0.35 $\pm$ 0.04	11.01 $\pm$ 2.27 $\rightarrow$ 7.37 $\pm$ 2.63
RESM-faithful	1.80 $\pm$ 0.09 $\rightarrow$ 0.32 $\pm$ 0.11	22.20 $\pm$ 0.16 $\rightarrow$ 11.12 $\pm$ 1.54	35.74 $\pm$ 0.22 $\rightarrow$ 17.93 $\pm$ 3.64	0.64 $\pm$ 0.65 $\rightarrow$ 0.38 $\pm$ 0.66	0.46 $\pm$ 0.01 $\rightarrow$ 0.35 $\pm$ 0.03	11.12 $\pm$ 2.66 $\rightarrow$ 7.53 $\pm$ 2.92
single-run reference rows (no seed band; pre $\rightarrow$ post only)						
RESM-faithful @10 ⁻⁴	2.23 $\rightarrow$ 0.60	22.88 $\rightarrow$ 6.72	34.98 $\rightarrow$ 16.21	1.27 $\rightarrow$ 0.22	0.32 $\rightarrow$ 0.17	12.00 $\rightarrow$ 8.11
RESM-released [§]	1.730 $\pm$ 0.005 $\rightarrow$ 0.203 $\pm$ 0.008	22.42 $\rightarrow$ 10.71 $\pm$ 0.27	35.26 $\rightarrow$ 15.29	0.97 $\rightarrow$ 0.20 $\pm$ 0.00	0.42 $\rightarrow$ 0.33 $\pm$ 0.00	12.11 $\rightarrow$ 9.10
RESM @10 ⁻⁴ , 30k	2.08 $\rightarrow$ 0.36	23.14 $\rightarrow$ 6.12	35.76 $\rightarrow$ 13.88	1.92 $\rightarrow$ 1.64	0.29 $\rightarrow$ 0.18	11.94 $\rightarrow$ 8.27